THE UNIVERSITY OF HONG KONG

MASTER THESIS

# From Medical Device Innovation to a Novel Tendon-Driven Mechanism: The Design and Validation of Lasso Gripper

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Abstract of thesis entitled

# From Medical Device Innovation to a Novel Tendon-Driven Mechanism: The Design and Validation of Lasso Gripper

Submitted by

**Yu WANG**

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The newly designed laparoscopic instrument employs two motors and two magnetic sensors as its input system, along with four motors dedicated to output actuation. This configuration is meticulously optimized to maintain continuous cable tension, ensuring consistent force transmission and precise control during operation. By integrating advanced filtering and control algorithms, the instrument effectively minimizes the impact of external disturbances and unwanted vibrations. As a result, it accurately translates the user's input gestures into smooth and responsive motions at the end-effector, enabling five essential degrees of freedom: rolling, yawing, pitching, and opening/closing of the jaws. This enhanced maneuverability and stability make the instrument particularly suited for delicate and complex minimally invasive surgical tasks.

The cable management system plays a critical role in sustaining tension across all driven tendons, thereby avoiding slack and enhancing motion accuracy. The inclusion of real-time sensor feedback allows for closed-loop control, which further refines the instrument's responsiveness and reliability. Such a design not only reinforces operational safety but also contributes to the system's adaptability in various surgical scenarios.

Inspired by the tendon-driven mechanism, Lasso Gripper is design and manufactured to grasp and manipulate oversized, variably-shaped, or delicate objects. This remains a significant challenge in robotics, largely due to the inherent limitations in the geometry and actuation mechanisms of conventional grippers. Drawing inspiration from traditional tools such as the lasso and the uurga, this paper presents Lasso Gripper—a novel end-effector that captures objects by launching and retracting a string loop. Unlike antipodal grippers, which apply concentrated force on limited contact areas, Lasso Gripper distributes pressure uniformly along the string, enabling a gentler and more adaptive grasp suitable for fragile and irregularly shaped targets.

The system is driven by four motors: two dedicated to reeling the string in and two for launching it outward. By independently regulating motor speeds, the size of the loop can be dynamically adjusted to accommodate objects of various dimensions, effectively overcoming constraints related to maximum gripper opening width. A specially designed mechanism is incorporated to prevent string tangling during high-speed retraction, enhancing reliability during operation. Furthermore, a dynamic model of the string is developed to estimate its configuration in real-time, facilitating kinematic analysis and workspace estimation.

In experimental trials, Lasso Gripper, mounted on a robotic arm, successfully captured and transported a diverse set of objects—including animal figures such as bulls and horses, as well as delicate vegetables—demonstrating its versatility and robustness. The results confirm its superiority over traditional grippers in handling a wide spectrum of objects with minimal prior knowledge of their geometry or fragility.

Beyond grasping applications, this research extends into medical technology by deconstructing and reconfiguring cable-driven mechanisms. The principles derived from Lasso Gripper are applied to the design and control of a new motorized laparoscopic instrument.

The newly designed laparoscopic instrument employs two motors and two magnetic sensors as its input system, along with four motors dedicated to output actuation. This configuration is meticulously optimized to maintain continuous cable tension, ensuring consistent force transmission and precise control during operation. By integrating advanced filtering and control algorithms, the instrument effectively minimizes the impact of external disturbances and unwanted vibrations. As a result, it accurately translates the user's input gestures into smooth and responsive motions at the end-effector, enabling five essential degrees of freedom: rolling, yawing, pitching, and opening/closing of the jaws. This enhanced maneuverability and stability make the instrument particularly suited for delicate and complex minimally invasive surgical tasks.

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(464 words)

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A Thesis Submitted in Fulfilment
of the Requirements for the Degree of
Master of Philosophy

at

The University of Hong Kong
September, 2025

# Declaration

I, Yu WANG, declare that this thesis titled, "From Medical Device Innovation to a Novel Tendon-Driven Mechanism: The Design and Validation of Lasso Gripper", which is submitted in fulfillment of the requirements for the Degree of Master of Philosophy, represents my own work except where due acknowledgement have been made. I further declared that it has not been previously included in a thesis, dissertation, or report submitted to this University or to any other institution for a degree, diploma or other qualifications.

Signed: Yu WANG

Date: September 29, 2025

*For Mama and Papa*

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Yu WANG
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# List of Publications

## JOURNALS:

[1] R Peng, **Y Wang**, M Lu, P Lu, “A dexterous and compliant aerial continuum manipulator for cluttered and constrained environments,” *Nature Communications*, January 2024.

[2] R Peng, **Y Wang**, P Lu, “A Tendon-Driven Continuum Manipulator With Robust Shape Estimation by Multiple IMUs,” *IEEE Robotics and Automation Letters*, 2024, 9(4), 3084 - 3091.

[3] W Guan, P Chen, H Zhao, **Y Wang**, P Lu, “EVI-SAM: Robust, Real-time, Tightly-coupled Event-Visual-Inertial State Estimation and 3D Dense Mapping,” *Advanced Intelligent Systems*, 2024, 1-24.2024, 1-24..

## CONFERENCES:

[1] **Y Wang***, Q Qiao*, X Fan, P LU, "Lasso Gripper: A String Shooting-Retracting Mechanism for Shape-Adaptive Grasping", *IEEE International Conference on Robotics and Biomimetics (IEEE ROBIO 2025)*, August 2025, Under Reivew.

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# Chapter 1

# Introduction

In laparoscopic surgery, surgeons are required to manipulate long, rigid instruments through small incisions, which severely restricts their degrees of freedom and attenuates tactile feedback, shown in Fig.1.1. Conventional laparoscopic tools typically offer only basic grasping and rotation, lacking the articulated movement of a human wrist. Inspired by the principles of Lasso Gripper, a compelling extension emerged: Could the "pull-release" philosophy of controlling a loop be deconstructed into the coordinated control of multiple tendons, and integrated into the distal end of a handheld instrument to restore wrist-like functionality for surgeons in a minimally invasive setting? Motivated by this vision, the latter part of this research is dedicated to the design, implementation, and evaluation of a novel, cable-driven, multi-DoF, and modular laparoscopic surgical instrument. The primary contribution of this research is the de-

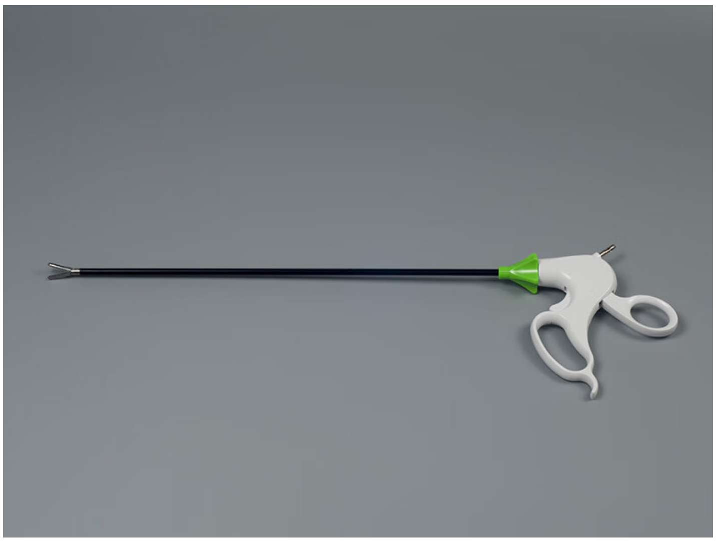

**Figure 1.1:** Typical Single Hole Surgical Instruments

velopment and extension of a cable-driven manipulation paradigm. It begins with the invention of a lasso gripper, whose controlled launch-and-retract mechanism provides a unique solution for adaptive and non-invasive grasping. Building on this foundation, the work makes a significant translational contribution by introducing a new handheld laparoscopic device. Leveraging similar cable-control theories, this instrument utilizes input and output motors to create a stabilized system that filters tremor and assists the surgeon in executing precise manipulations, thereby bridging a novel robotic grasping concept with a practical healthcare application.

In robotics, cable-based mechanisms are seldom employed in primary design applications due to their inherent lack of rigidity, which often limits positional accuracy and load-bearing capacity. Even in scenarios requiring flexible power transmission, toothed belts or timing belts are generally preferred for their positive engagement, minimal slippage, and precise motion control enabled by grooved profiles. However, in the specific domain of grasping—particularly for unstructured, delicate, or variably shaped objects—cables offer unique functional advantages. Their exceptional flexibility allows them to passively conform to the geometry of both the gripper and the target object, enabling a form-fitting, enveloping grasp that is difficult to achieve with rigid or semi-rigid components. This adaptive capability makes cable-driven systems highly suitable for applications where gentle yet secure handling is paramount.

the capability to swiftly and securely grasp and manipulate objects is crucial, especially in environments that demand high-speed operations and substantial load-bearing capacities. Traditional robotic grippers often encounter limitations in either operational speed or load capacity, reducing their efficiency in dynamic and challenging applications [10]. As technological advancements progress, the need for more versatile and effective gripping solutions becomes increasingly pressing.

This essay investigates the design and underlying principles of a groundbreaking "Lasso Gripper," a mechanism inspired by traditional tools like the lasso and the uurga. Lasso Gripper is devised to surmount the constraints of existing grippers by merging a broad operational workspace with substantial load capacity and adaptability to various object shapes. This innovative design aims to bridge the gap between the flexibility of pneumatic grippers and the extended reach of flying claw mechanisms.

Lasso Gripper offers several significant advantages over traditional grippers. One of the primary benefits is its suitability for grasping high-speed objects. The string loop can ensnare objects within a larger capture range, allowing Lasso Gripper to have resistance to uncertainties of the object's position. This means that the gripper can accommodate errors in position estimations at the moment of grasping, increasing its effectiveness under dynamic situations.

Additionally, Lasso Gripper is well-suited for handling heavier loads because the string loop applies tensile force to the object, unlike parallel-finger grippers that rely purely on friction to hold objects. This fundamental difference in gripping mechanism allows Lasso Gripper to secure heavier objects more reliably. Moreover, the use of wool or nylon materials for the string imparts a softness that enhances shape adaptability, making Lasso Gripper similar to other flexible robotic hands in this respect.


**Figure 1.2:** Traditional cowboy lasso

**Figure 1.3:** Traditional Mongolian uurga

The development of Lasso Gripper is driven by the necessity to combine the best features of current gripping technologies. By utilizing a string loop that can be rapidly shot and retracted, Lasso Gripper ensures a secure grasp without slippage, even in high-speed and high-load scenarios. Drawing inspiration from the traditional lasso used by cowboys and the uurga employed by Mongolian herders, the gripper leverages the mechanical advantages of a string loop to enhance its gripping effectiveness.

The key to Lasso Gripper's design is its ability to achieve a large initial capture range, significantly increasing the likelihood of successful grasping, particularly with fast-moving targets. This essay will explore the detailed design and operational principles of Lasso Gripper, including its overall structure and the specifics of the string shooting-retracting mechanism.

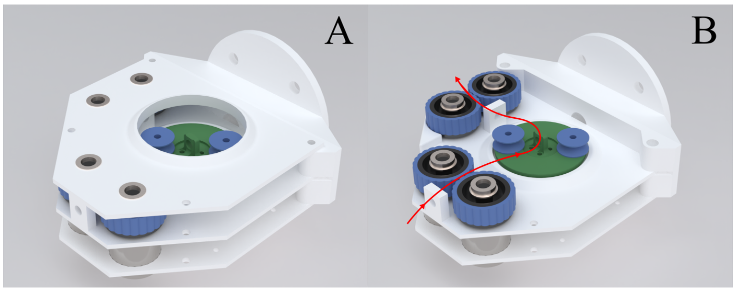


**Figure 1.4:** The prototype of proposed Lasso Gripper

In addition, the essay will discuss the innovative design features that set Lasso Gripper apart from existing technologies. By evaluating experimental results and comparing Lasso Gripper's performance with other gripping mechanisms, this essay aims to highlight the advantages of Lasso Gripper in various scenarios. Ultimately, Lasso Gripper represents a significant advancement in robotic gripping technology, offering a robust solution for rapid and heavy-duty object manipulation.

The successful design of Lasso Gripper demonstrates the significant potential of cable-driven actuation and grasping strategies in unstructured environments. However, the ambition of this research extends beyond the development of a novel robotic gripper. It has been recognized that the core advantages of cable-driven mechanisms — namely, their inherent compliance, the ability to locate actuators remotely, and the flexibility in decoupling complex motions — directly address a long-standing challenge in another field: the lack of dexterity in Minimally Invasive Surgery (MIS), particularly in laparoscopic procedures.

# Chapter 2

# Related Works

## 2.1 Advances in Actuation Technologies for Multi-DoFs End-Effectors in Laparoscopic Instruments

Prior to the advent of Minimally Invasive Surgery (MIS), conventional abdominal procedures typically required a large, open incision—often referred to as a laparotomy—to provide surgeons with direct visual and physical access to the target anatomy. The instruments employed in such open surgeries were generally limited to a single DoF, such as simple grasping or cutting, while the remaining necessary dexterity — including rotation, flexion, and extension — was supplied entirely by the surgeon's own wrist and arm movements. Although effective in an open setting, this approach resulted in significant tissue trauma, prolonged patient recovery, and increased postoperative pain. In response to these limitations, the development of MIS spurred the exploration of multiple technological approaches to restore dexterity and precision within the constrained environment of keyhole surgery.

To address the limited dexterity at the instrument tip, a prominent solution involves using cables as a medium to transmit power from the handpiece to the end-effector, thereby enabling enhanced DoFs, such as roll or yaw rotations. First introduced in the late 1990s, Intuitive Surgical's da Vinci system exemplifies this approach, employing a cable-driven mechanism to control its articulated tips. The primary advantage of such a system is that cables can transmit power over a distance with a lower weight penalty compared to rigid linkages or gear trains[2]. However, this benefit is accompanied by inherent challenges. In the initial design of the da Vinci system, cable coupling—where the movement of one cable inadvertently affects another—was a significant issue that could not be overlooked, often leading to kinematic inaccuracies and reduced control fidelity. Therefore, various approaches have been studied to address these problems.

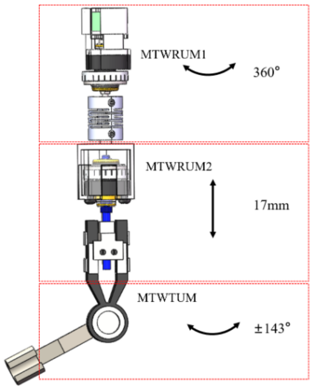


**Figure 2.1:** Design and Illustration of Ultrasonic Motors[15].

### 2.1.1 Replacements of Driven Motors

To address the challenges of control fidelity loss and restricted tip articulation in minimally invasive surgical instruments, various driving mechanisms have been extensively investigated. One prominent approach involves the implementation of ultrasonic motors as alternatives to conventional electromagnetic actuators. As illustrated in Fig. 2.1, experimental validation has confirmed that the maximum output torque reaches 1.2 mN·m for the rotary configuration and 1.4 mN·m for the tilting mechanism. These torque values adequately satisfy the operational requirements of magnetically anchored laparoscopes while offering superior positional precision. Crucially, performance evaluations of the prototype system have substantiated both the feasibility and effectiveness of integrating these ultrasonic motors into laparoscopic applications[13]. This approach presents a novel and practical alternative for achieving enhanced visual platform maneuverability in single-port surgery, addressing fundamental limitations in conventional cable-driven systems through improved torque transmission and positioning accuracy.

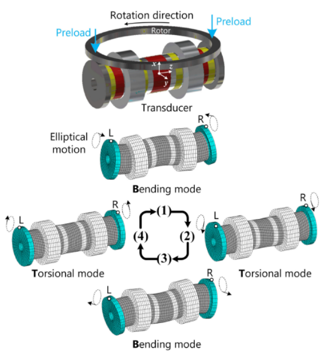


**Figure 2.2:** Bending and Torsion layouts of the Transducer[33].

To match the power density and peak torque output of this proposed driving method to electromagnetic actuators, there are studies that focus on this area. By applying a rod-shaped transducer in torsional and bending modes, the rotor is driven in its orthogonal direction with external excitements at certain frequencies. By adjusting exciting frequencies, the maximum torque recorded in the experiments reaches 10.1 N.m and output power reaches 38.1 W[33].

### 2.1.2 Additional Sensors

A fundamental limitation in the original cable-driven system involves the progressive uncertainty of cable length resulting from material creep during prolonged operation. This phenomenon introduces cumulative positioning errors between the input and output ends, significantly compromising control fidelity[19]. To mitigate this inherent uncertainty, the integration of supplemental sensing modalities has been implemented. These sensors enable real-time monitoring of either cable tension, end-effector pose, or both, facilitating a closed-loop control architecture that dynamically compensates for length variations and restores positional accuracy.

Accurate measurement of cable tension during surgical procedures represents a critical safety requirement in cable-driven robotic systems. In traditional laparoscopic surgery, surgeons rely on haptic feedback transmitted through rigid instruments to assess tissue interaction forces. However, in motor-driven systems where actuation forces are generated remotely, this crucial tactile information becomes significantly attenuated. This sensory disconnect necessitates extensive training for surgeons to develop appropriate motor control through the handheld interface, as demonstrated in studies on force sensing in robotic surgery [30].

Force sensing has emerged as a promising solution to restore this lost feedback. The most straightforward approach involves embedding force sensors, such as Fiber Bragg Gratings (FBGs), directly into the surgical instrument. As explored in research on mechatronic design of force sensors[38], this method enables direct measurement of interaction forces at the end-effector. However, this integration often results in structural redundancy and increased instrument diameter, creating bulky designs that may compromise surgical access and maneuverability in confined operative fields.

To address these limitations, recent research has focused on miniaturized sensing strategies. One innovative approach involves embedding FBGs within precisely machined grooves on inclined cantilevers within the instrument structure[35]. This design exploits the strain measurement capabilities of FBGs to detect minute deformations in the cantilevers caused by cable tension variations. The inclined configuration amplifies these deformations, enabling high-sensitivity force measurement while maintaining a compact form factor compatible with the dimensional constraints of minimally invasive instruments.

### 2.1.3 Novel Structures of the Tip and Rod Design

In conventional master-slave robotic systems, such as the da Vinci Surgical System and similar handheld robotic instruments, the end-effector typically provides 4 to 5 DoFs. This configuration is designed to replicate the articulation capabilities of the human wrist and fingers, enabling complex maneuvers like pitching, yawing, rolling, and grasping. However, a significant design limitation persists: these multiple DoFs are concentrated exclusively at the distal tip, while the connecting shaft or linkage primarily functions as a passive rotary base.

This recognition of ergonomic limitations in conventional articulated instruments has stimulated significant research into alternative tip architectures in recent years. With advances in the modeling and control of continuum robots, their design principles have been increasingly translated into the medical domain. Unlike traditional tip designs composed of multiple rigid links connected by discrete joints, continuum robots offer a validated scheme for implementing soft or continuously flexible connections—both between the tip and the instrument shaft, and among individual subcomponents of the tip itself. This paradigm shift enables a more distributed deformation along the instrument's length, effectively increasing the overall dexterity without concentrating all articulation at a single point. The resulting designs, often drawing inspiration from biological structures like tentacles or snakes, promise to mitigate surgeon fatigue by decoupling the end-effector's pose from the precise position of the surgeon's hand, thereby offering a more intuitive and ergonomic manipulation platform for complex surgical tasks. Building upon the foundational research in contin-

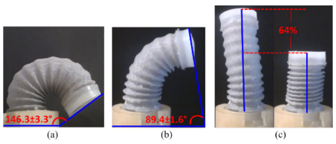


**Figure 2.3:** flexible fluidic actuators formed with granular to vary stiffness [24]

uum robotics, recent studies have explored innovative structural designs to enhance surgical instrument dexterity and safety. Ranzani et al. (2016) demonstrated a significant advancement by integrating flexible fluidic actuators within a silicone matrix containing multiple pneumatic chambers. This design enabled multi-directional bending and elongation capabilities. The proposed manipulator employed a modular architecture, utilizing granular jamming techniques to achieve dynamic stiffness variation and diverse movement profiles, as illustrated in 2.3. Further extending this paradigm, researchers have investigated the transition from rigid linkages to compliant structures to substantially improve operational flexibility. One notable approach involves manipulating the trimmed angle of a helicoid structure to create a variable-stiffness robotic arm[9]. When coupled with a cable-driven actuation system, this design achieves an

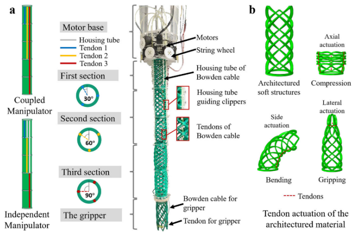


**Figure 2.4:** The tendon-driven manipulator design with Trimmed Helicoids[9]

expanded workspace while significantly minimizing the risk of iatrogenic tissue damage during surgical procedures, as shown in 2.4. These developments in soft robotics and variable-stiffness mechanisms represent promising directions for next-generation surgical instruments that balance precision with patient safety.

### 2.1.4 AI-assisted Monitor and Control

As analyzed in previous sections, contemporary research in medical instrumentation predominantly advances along three interconnected fronts: innovative tip design, integrated sensory feedback systems, and intelligent control algorithms. The rapid evolution of Deep Learning and Reinforcement Learning methodologies has now catalyzed a fourth research direction: the integration of data-driven approaches into clinical practice.

A particularly promising development lies in repurposing the existing endoscopic camera—initially intended solely for surgical visualization—as an external verification system for instrument navigation. Specifically, Kong et al. (2023) demonstrated that through a specially designed architecture, both the visual stream from the camera and kinematic data from the actuation motors can be fused to estimate the positional discrepancy between commanded and actual end-effector positions following movement execution[14]. This estimated deviation is subsequently converted into compensation signals that drive automatic self-correction maneuvers, effectively creating a vision-enhanced closed-loop control system that continuously calibrates the instrument without requiring additional hardware beyond the standard surgical setup.

This methodology represents a significant paradigm shift from traditional model-dependent control strategies, leveraging the wealth of available surgical video data to address inherent system limitations such as cable stretch, hysteresis, and kinematic modeling inaccuracies. The approach demonstrates how computer vision and deep learning techniques can transform existing surgical infrastructure into a self-calibrating system, potentially enhancing both positioning accuracy and long-term reliability in cable-driven surgical instruments.

## 2.2 Development of Adaptive Grasping and its application

### 2.2.1 The Necessary of Enveloping and Encircling Grasping

The necessity of enveloping and encircling grasping techniques in robotic technology is evident due to their adaptability to diverse objects and enhanced grasping stability. Recent advancements in this field demonstrate various approaches to achieving these objectives.

One notable study introduces a dual-mode soft robotic gripper designed with integrated annular structures and a contraction actuator. This design significantly improves shape tolerance and gripping force by altering the contact method between the gripper and the object, adjusting the stiffness and load capacity through controlled material properties [11].

Another novel technique employs a soft gripper that leverages a belt loop with an actuated adhesion mechanism. This mechanism provides dual-directional force – both radial and tangential – which enhances the effectiveness and stability of the grasping action. The minimal interaction force strategy ensures gentle handling, making it suitable for fragile objects [4].

Additional studies introduce an affordable, underactuated compliant gripper capable of adapting to objects of diverse sizes and shapes. The design features a flexure motion transmission mechanism and multifunctional jaws, allowing for a more conforming grasp. This interaction between the gripper and the target object increases both gripping force and stability [27].

Additionally, a study on a tendon-driven robotic gripper highlights the importance of stable enveloping grasps. By optimizing tendon routes and stiffness, this gripper can perform multiple grasping modes, including parallel, enveloping, and fingertip grasps. This versatility ensures the gripper's flexibility and stability across various applications [5].

Enveloping and encircling grasping techniques are crucial in robotic grasping technology. They enhance shape adaptability, gripping force, and stability, providing robust support for robots in complex and dynamic environments.

### 2.2.2 Grasping System with Integrated Sensors

The integration of sensors with soft materials such as loops significantly enhances the efficiency and adaptability of robotic grasping systems. Various studies have demonstrated the potential benefits of combining these elements to improve performance in different contexts.

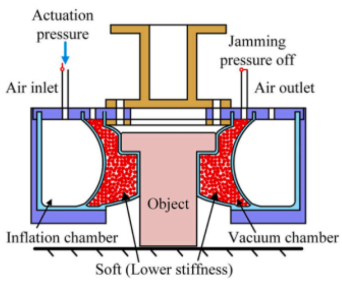


**Figure 2.5:** Pressurized the actuation chamber and adaptive grasping [11]

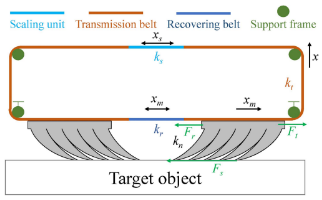


**Figure 2.6:** Belt Loop Actuated Adhesion Design [4]

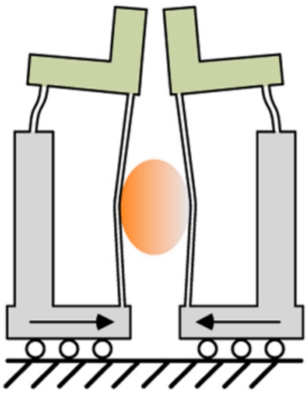

**Figure 2.7:** Enveloping grasps of the multi-functional jaw [27]

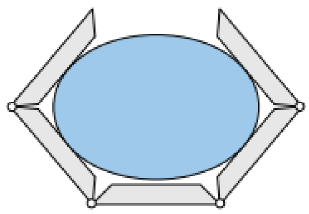

**Figure 2.8:** Examples of enveloping grasps of Velo gripper [5]

A notable example is a modular flying gripper that utilizes a loop gripper mechanism to control its position and pose flexibly. This system, composed of quadrotor-based modules, showcases the versatility of encircling grasping techniques. The decentralized control approach enables the flying gripper to modify its aperture angle, thereby facilitating efficient object grasping and transportation [8].

In another study, a soft robotic gripper equipped with a liquid-metal sensing skin provides a robust feedback mechanism. This gripper, integrated with proximity, pressure, and orientation sensors, operates using shape-memory alloy springs to transition between open and close states. The closed-loop control enabled by these sensors allows the system to detect and respond to external disturbances, enhancing its grasping, release, and transportation capabilities [36].

Further advancements are seen in the development of a continuum body gripper that incorporates flexible latex bladder sensors. By measuring air pressure changes, these sensors provide tactile and proprioceptive feedback, essential for delicate and high-force grasping. This integration allows the gripper to manage a diverse array of objects, ranging from heavy to fragile items, with minimal control input. The sensorized gripper demonstrates high sensitivity and repeatability, making it a cost-effective and durable solution for universal grasping applications [12].

Additionally, a visual-tactile fusion framework has been proposed to address the challenges of grasping transparent objects in complex environments. This framework combines visual and tactile data to enhance grasping efficiency. A tactile calibration technique combined with a visual-tactile fusion classification method greatly enhances the success rate and accuracy of grasping transparent objects. The experimental results validate the effectiveness of integrating tactile sensors within the gripper system, particularly for objects that are irregular or visually undetectable [17].

The combination of sensors with soft materials such as loops in robotic grippers offers significant improvements in grasping systems. These integrations provide enhanced feedback, adaptability, and control, rendering them invaluable for a wide array of applications, from delicate object handling to complex environmental interactions.

**Figure 2.9:** The Flying Gripper holding a coffee cup [8]

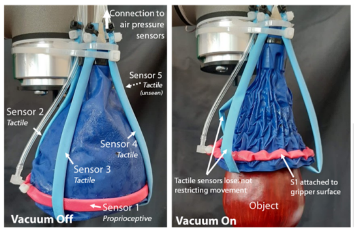


**Figure 2.10:** The illustration of tactile sensor and pressure sensors integrated in the loop gripper [12]

### 2.2.3 Designing for the Physical Properties of Soft Gripper

Through advancements in material technology and microstructural design, significant progress has been made in altering the physical properties of flexible materials like ropes. These innovations have paved the way for the development of more efficient and adaptable soft robotic grippers.

A groundbreaking soft gripper utilizing a self-sensing super-coiled polymer (SCP) actuator is introduced by Wang et al. (2021). This actuator is crafted from multi-strand sewing thread that undergoes a self-coiling process, enabling it to untwist when heated. The addition of a metal wire wrapped around the SCP enhances temperature measurement precision, forming a closed-loop control system for accurate gripper manipulation. This approach offers a foundation for customizing the physical properties of rope-like materials for actuator applications [31].

Sun et al. (2020) described a soft gripper with adjustable stiffness, inspired by the behavior of pangolin scales. The gripper is composed of two layers: a variable stiffness layer and a pneumatic-driven layer. The variable stiffness layer emulates pangolin scales, remaining flexible during normal activities but becoming rigid when threatened. The toothed pneumatic actuator increases the gripper's stiffness, enabling it to grasp a wide range of objects effectively. The autonomous control system of this gripper collects feedback signals to adjust the stiffness autonomously, making it suitable for unstructured environments [26].

Zhang et al. (2017) tackled the design problem of soft robots using a topology optimization framework. A pneumatically actuated soft gripper was developed, with three fingers each designed as a continuum compliant mechanism to achieve maximum bending deformation. This actuator was fabricated through 3D printing, and experimental results demonstrated its ability to grasp objects effectively. This optimization approach provides valuable insights for further exploration of the microstructure of lasso grippers [39].

Li et al. proposed a soft gripper inspired by the inchworm's true leg. This design includes two rows of sharp hooks on a soft palm, with connections made via Shape Memory Alloy (SMA) springs. The design allows the gripper to adhere to rough surfaces and grasp objects with varying shapes and dimensions. The combination of elastic muscle-like structures and sharp hooks enhances the load-carrying capacity of the gripper, demonstrating the potential of microstructural design in soft robotics [16].

Wang et al. (2017) explored the usage of Shape Memory Alloy (SMA) in soft grippers, addressing the inherent softness and low actuation force of traditional systems. The study described a soft gripper with variable stiffness based on SMA, composed of three fingers, each with two hinges. The hinges can achieve a significant change in stiffness, allowing the gripper to adapt to various shapes and weights. This material modification approach shows promise for effective grasping applications [32].

The integration of advanced material technologies and microstructural designs has significantly enhanced the capabilities of soft robotic grippers. These innovations provide a robust foundation for developing adaptable and efficient grasping systems.

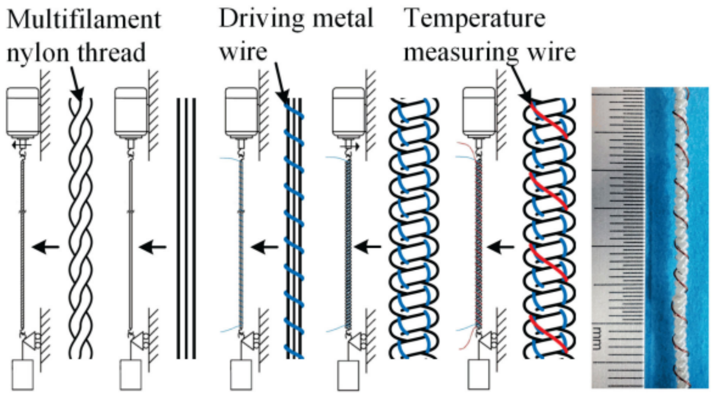


**Figure 2.11:** Manufacturing process of actuator with multifilament nylon thread [31]

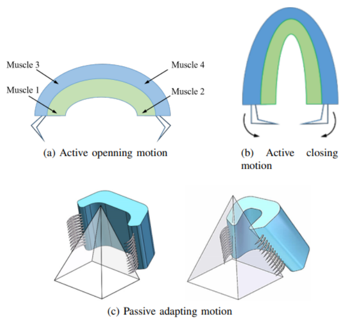


**Figure 2.12:** Bio-mechanism of inchworm's true leg [16]

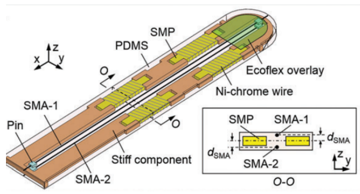


**Figure 2.13:** Shape Memory Alloy-Based Soft Gripper [32]

### 2.2.4 Advancement of Grasp Planning Algorithm for Lasso Grippers

The design of grasping algorithms has made significant strides in enhancing the adaptability and efficiency of robotic grippers, particularly those employing rope-like structures such as lasso grippers. These algorithms leverage unique approaches to optimize grasp planning and ensure stable and robust manipulation.

Li et al. (2015) proposed an innovative grasp planning method that addresses the computational challenge of determining stable force-closure grasp configurations for multi-fingered hands. A low-level shape exploration technique is employed in this method by wrapping multiple cords around an object to quickly identify promising grasping regions. Using a variant of the close-until-contact procedure, contact points are determined by the algorithm, which then utilizes finger kinematics and contact information to filter out unstable grasps. This approach not only enhances the natural appearance of grasps but also improves efficiency, as it bypasses extensive preprocessing steps like segmentation and medial axis extraction. The adaptability of this method suggests that the structural concept of wrapping cords around objects can be effectively applied to lasso grippers, potentially offering superior grasping performance [18].

Fang et al. (2023) introduced the AnyGrasp algorithm, designed to enhance robust and efficient grasp perception across both spatial and temporal domains. This algorithm utilizes a dense supervision strategy, integrating real perception data with analytic labels to create a comprehensive training dataset. By incorporating object center-of-mass awareness into the learning process, AnyGrasp improves grasp stability. Additionally, it employs grasp correspondence across observations, enabling dynamic tracking of grasp poses, which ensures effectiveness even in noisy environments. The success of AnyGrasp in achieving a high success rate in bin clearing tasks underscores its potential applicability to various robotic grippers, including those with lasso structures [7].

These advancements in grasping algorithm design underscore the importance of innovative computational methods in enhancing the functionality of robotic grippers. The adaptive planning optimization demonstrated by Li et al. (2015) highlights the potential of rope-like structures in improving grasp performance. As the field continues to evolve, integrating these approaches with lasso grippers could lead to more versatile and effective robotic grasping systems.

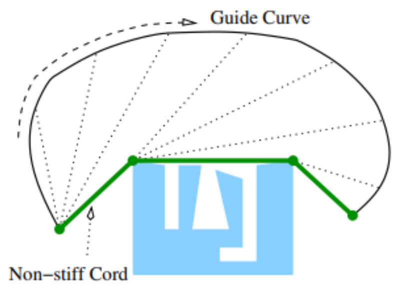


**Figure 2.14:** Geometric construction of a non-stiff cord [18]

### 2.2.5 Application of Lasso Grippers in Real-World Scenarios

The application of lasso grippers in various real-world scenarios demonstrates their versatility and effectiveness in handling a variety of objects with minimal control effort and high reliability. This section reviews recent advancements and practical implementations of lasso grippers across different fields.

Manes et al. (2024) introduced a soft cable loop-based gripper designed for robotic automation in chemistry laboratories. This gripper can reliably grasp cylindrical and prismatic objects of various sizes with minimal clearance. The design utilizes the compliance of a cable to completely envelop target objects, with support provided by a rigid finger once the grip is secured. This mechanism requires minimal control effort, making it highly practical for repetitive tasks in automated chemistry processes. The gripper demonstrated a grasp reliability with a failure rate of less than 1%, showcasing its efficacy in handling delicate laboratory glassware. However, the use of a self-supporting plastic polymer material limits the loop size, as larger loops would fail to maintain their shape [21].

**Figure 2.15:** A cable loop gripper prototype was evaluated for its ability to grasp various types of chemistry glassware. [21]

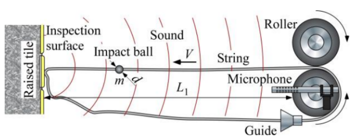


**Figure 2.16:** String shoot impact towards a wall [29]

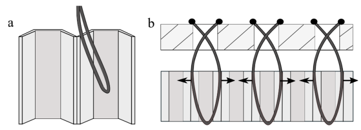


**Figure 2.17:** String loop gripper for honeycomb materials [25]

A cable-driven gripper with perception capabilities for autonomous strawberry picking was developed by Xiong et al. (2018). The gripper utilizes six fingers to create a closed space around a target strawberry, gently cutting the stem without causing damage to the fruit. Equipped with infrared sensors, positional errors can be corrected by the gripper, enhancing robustness and achieving a high success rate in picking isolated strawberries. The integration of internal perception allows the gripper to tolerate positional inaccuracies, reducing the need for slow, high-level closed-loop control. This design highlights the advantages of using cable actuators, such as flexibility and biological friendliness, making it suitable for delicate agricultural applications [34].

Roth et al. (2021) presented the Loop Gripper designed for handling honeycomb materials used in various industries. This gripper uses adjustable threads to provide the necessary flexibility and sensitivity to grip honeycomb panels without damaging their fragile cell walls. Through a series of tests, the optimal gripping parameters for various honeycomb materials were determined, proving the concept's suitability for automated gripping tasks. The Loop Gripper's approach of using a soft, structure-friendly force to grasp objects aligns with the principles of lasso grippers, emphasizing the importance of adaptability and gentle handling [25].

These studies underscore the practical applications of lasso grippers in diverse fields, from chemistry laboratories to agricultural and material handling tasks. The inherent adaptability and gentle handling capabilities of lasso grippers make them an effective solution for various automated processes, demonstrating their potential for widespread adoption in real-world scenarios.

### 2.2.6 The Balance between Flexibility and Strength

Current research in robotic gripping mechanisms often focuses on achieving a balance between flexibility and strength. Pneumatic flexible grippers, for instance, offer excellent shape adaptability but are limited in their working range and load capacity. Yu et al. have noted that soft pneumatic actuators are widely utilized in soft robots, where their bending angles and motion rules at different pressures are crucial for practical applications [37]. However, despite their flexibility, these actuators struggle with heavier loads and larger workspaces. Moreover, their dependency on precise pressure control can make them susceptible to performance inconsistencies, particularly in environments where maintaining stable pressure is challenging.

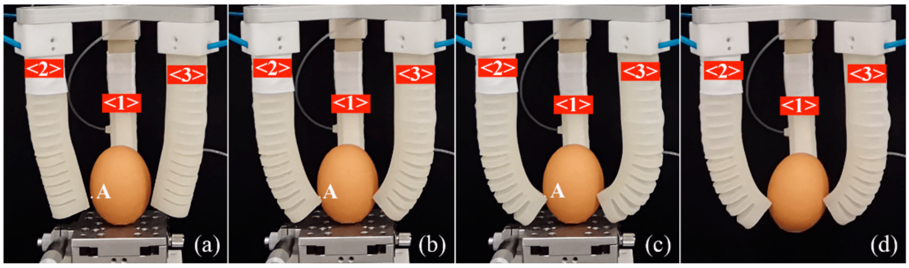


**Figure 2.18:** Pneumatic actuator grasping process of an egg [37]: (a) The soft actuator moves toward the egg when pneumatic module one is pressurized to 10 kPa, with module two remaining at zero pressure. (b) Contact between the tip of the soft actuator and the egg occurs when the pressure in module one is increased to 20 kPa and module two is set to 14.1 kPa. (c) The egg is securely pinched by the gripper when both module one and module two are pressurized to 20 kPa. (d) The egg is held firmly when the supporting platform is removed, indicating a stable grasp by the gripper.

On the other hand, Chameleon-like shooting manipulator offer extended working ranges but suffer from lower success rates and reduced effective working ranges [22]. Such systems often rely heavily on complex control algorithms and precise timing, which can be difficult to manage in real-time applications, leading to potential delays and reduced effectiveness in fast-paced environments.

Fractal grippers, as discussed by Tisdale and Burdick, draw inspiration from an older concept called the Fractal Vise [28]. Their work emphasizes the gripper's ability

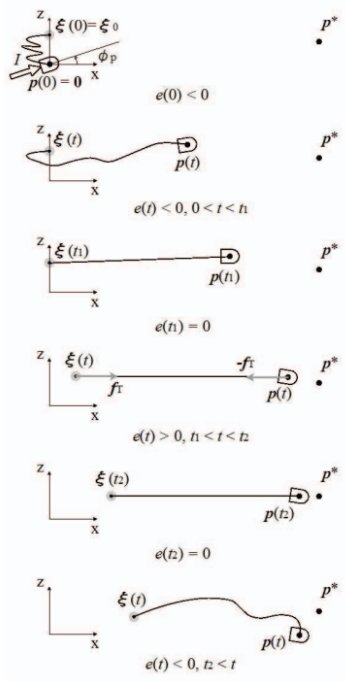


**Figure 2.19:** Manipulating model of chameleon-like gripper [22].

to conform passively to various objects using a single actuator. Nonetheless, while these mechanisms offer synergistic properties and adaptability, their integration with existing systems and capacity for high-load tasks are still limited. The passive nature of these grippers, while beneficial in certain contexts, can also lead to a lack of control over the gripping force, making them unsuitable for tasks requiring precise manipulation of heavy or delicate objects.

Lasso Gripper is designed to address these gaps by incorporating a mechanism that can rapidly shoot and retract a string loop, enabling it to capture objects with high speed and precision. Inspired by the traditional lasso used by cowboys and the uurga used by Mongolian herders, this mechanism leverages the structural advantages of a string loop to prevent slippage and ensure a secure grasp. Moreover, Lasso Gripper's design allows for a large initial capture range, significantly increasing the probability of successful grasping, particularly with fast-moving targets.

Brun et al. provide an insightful analysis of the mechanics of the lasso, illustrating how the rope's shape remains stable in the rotating reference frame of the hand, with line tension balanced by centrifugal force and the rope's weight [3]. This fundamental understanding of lasso mechanics informs the design and functionality of Lasso Gripper, ensuring that it can operate effectively under dynamic conditions.

This research will delve into the design and working principles of Lasso Gripper,

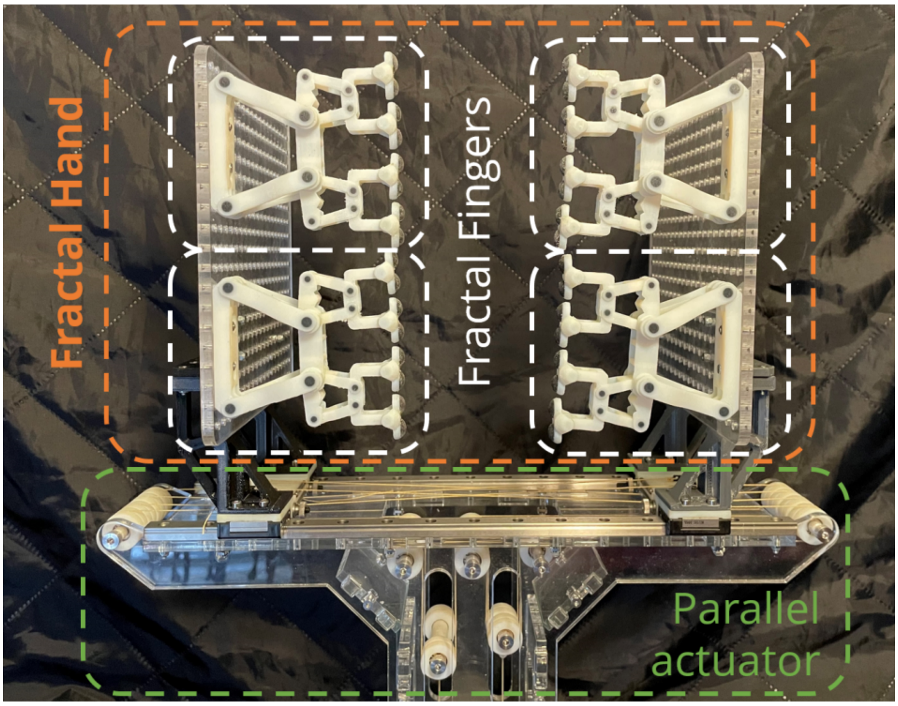


**Figure 2.20:** Fractal Hand prototype [28].

discussing its overall structure, the specifics of the string shooting-retracting mechanism, and additional design innovations within the gripper. We will also explore the grasping algorithm, which utilizes depth camera scanning and point cloud analysis to accurately locate and secure the target object. Furthermore, we will design experiments comparing Lasso Gripper's performance with other gripping mechanisms, highlighting its advantages in various scenarios.

# Chapter 3

# Innovative Design of a Handheld Surgical Instrument Balancing Weight and Functionality

## 3.1 Necessity To Develop Handheld Motorized Medical Instrument

As established in prior discussions, contemporary research in surgical robotics has predominantly prioritized maximizing tip-positioning accuracy through advanced actuation methods and the integration of supplementary sensors. While these approaches demonstrably enhance precision, they often overlook a critical practical constraint: the weight-to-functionality ratio. The evident correlation between enhanced capabilities—achieved through more powerful motors or additional sensing—and increased instrument mass creates a significant adoption barrier. This added weight often leads surgeons to favor simpler, nearly weightless traditional instruments over their motorized counterparts, despite the latter's technological advantages.

To resolve this fundamental trade-off, the present work introduces a holistic approach to ergonomic and functional design in robotic-assisted surgery. This chapter details the conception, design, and implementation of a novel structured handheld instrument that strategically balances mass, functionality, and clinical usability. The proposed system embodies a dedicated effort to distribute system components intelligently, minimize inertial load, and integrate multi-modal sensing without compromising the manipulator's agility or the surgeon's comfort. By fundamentally re-evaluating the instrument architecture from a human-centric perspective, this research aims to deliver a platform that surgeons are not only able to use, but genuinely prefer to use, thereby bridging the gap between technological potential and clinical practicality.

## 3.2 Core Design Advantages for Hand-Held Applications

This cable-driven, differential actuation philosophy offers several fundamental advantages that are particularly suited to the development of advanced hand-held devices:

-**Enhanced Dexterity and Manipulation**: The system enables multi-degree-of-freedom manipulation (e.g., rotation, twisting) using a minimal number of actuated elements. This is crucial for applications requiring fine motor control within confined spaces.

-**Inherent Compliance and Safety**: The system is naturally compliant. Cables under tension provide a direct force-feedback path. If an unexpected obstacle is encountered, the cables will slacken rather than apply a rigid, potentially damaging force. This passive safety feature is a critical asset for interactions in unstructured or delicate environments.

-**Proximal Actuation and Minimal End-Effector Mass**: A primary benefit for hand-held device design is the ability to locate the heavy actuators (motors) and associated drive electronics proximally, either in the device's handle or even in a remote unit. This dramatically reduces the mass and inertia of the distal end-effector, leading to improved ergonomics, reduced user fatigue, and the potential for higher maneuverability and smaller form factors at the interaction point.

## 3.3 Defining Performance Metrics for Structural Design

To translate this conceptual framework into a viable product, key performance metrics must be established. These metrics will directly guide the structural design choices detailed in the subsequent chapter:

-**Precision and Resolution**: The minimum achievable incremental movement of the end-effector tip, targeting sub-millimeter and sub-degree accuracy for delicate tasks.

-**Force Capability**: The maximum tensile force the cables must transmit to securely grasp and manipulate objects of a target weight range.

-**Range of Motion**: The required angular deflection and rotational range of the end-effector to perform its intended functions effectively.

-**Force Transmission Efficiency**: Minimizing friction losses throughout the cable drive path is essential for accurate force feedback and predictable control.

-**Low Hysteresis and Backlash**: The structural design must ensure minimal play and elastic deformation in the system to guarantee precise, repeatable, and immediate response to actuator inputs.

## 3.4 Pathway to Structural Implementation

This chapter has outlined the conceptual leap from a simple loop to a differentially actuated cable-driven system, highlighting its inherent advantages for dexterous manipulation in hand-held applications. The core principles of compliance, proximal actuation, and differential control form the foundation upon which a physical device must be built.

The following chapter will address the critical task of transforming this concept into a robust mechanical reality. The structural design must meticulously integrate several key subsystems:

1. **Actuation Unit Selection and Layout**: Choosing motors and sensors that meet the performance metrics and determining their optimal arrangement.
2. **Cable Routing and Guidance System**: Designing low-friction, precise pathways for the cables to ensure efficient force transmission and longevity.
3. **End-Effector Configuration and Cable Termination**: Engineering the interface where the cables attach to the manipulator, ensuring secure anchoring, precise motion, and mechanical advantage.

Through this structured approach, the subsequent design will validate the conceptual advantages discussed herein and realize a functional, high-performance cable-driven hand-held device.

## 3.5 Mechanism and Design

### 3.5.1 Output Motors

A paramount challenge in the design of handheld cable-driven devices lies in the selection of the actuation unit. The motor must satisfy two critical, and often conflicting, requirements: it must be extremely lightweight to maintain a low overall device mass and ensure ergonomic usability, while simultaneously being capable of delivering sufficient torque to reliably tension the driving cables for both grasping and manipulation tasks. High torque output is essential to overcome friction in the cable pathways, generate the necessary gripping force, and provide adequate acceleration for precise movements.

After a comprehensive evaluation of available options, Feetech STS3032, a smart servo was selected for this application. This choice was driven by its exceptional ratio over performance-to-weight and integrated control architecture. Key factors influencing this decision include: The STS3032 servo weighs a mere 9 grams, yet it delivers a substantial stall torque of 4.5 kg·cm. This high torque output ensures that the servo can generate significant tensile force in the cables, even when operating through small-diameter pulleys or under load, making it ideally suited for the demands of a surgical gripper.Unlike traditional pulse-width modulation (PWM) servos that require a dedicated control wire for each unit, the STS3032 utilizes a serial bus communication protocol. This architecture allows multiple servos to be daisy-chained and controlled via a single data line, drastically reducing the number of wires running through the device's handle. This reduction in cabling is a significant advantage for miniaturization, improving flexibility, simplifying assembly, and enhancing overall system reliability.

Another advantage of the this motor is that, as a smart servo, the STS3032 provides built-in feedback on parameters such as position, speed, and load. This enables closed-loop control strategies, which are critical for achieving the high positional accuracy and force sensitivity required for precise manipulation. The ability to read the motor's load provides valuable data for implementing basic force sensing, which can be used to prevent overloading and enable interactive control.

### 3.5.2 Input Motors and Sensors

The control philosophy for this cable-driven handheld device is based on replicating the user's hand movements with high fidelity and translating them into precise motions of the distal end-effector. This requires a sophisticated input system that can accurately capture the intent of the user's fingers while providing sufficient force feedback and stability. For the input actuation, shown in Fig.3.1, the Feetech STS3009 servo was selected based on a different set of requirements. Weighing 40 grams and providing a stall torque of 0.3 Nm, this more powerful motor serves two essential functions in the input stage: it maintains mechanical equilibrium by resisting forces from the output tendons, thus offering a stable and responsive feel to the operator, and it helps


**Figure 3.1:** Feetech 3032 for output

filter out high-frequency physiological tremors or unintended jerks, ensuring that only deliberate commands are transmitted to the end-effector.

However, since the two input motors alone cannot fully capture all nuanced finger movements, a hybrid sensing strategy was implemented. Two AMS AS5600 magnetic rotary encoders were integrated into the input mechanism to directly and accurately detect the user's intent. These sensors provide ultra-high 12-bit resolution and contactless sensing, enabling precise recognition of two distinct actions: the pinching gesture of the index finger and thumb, which controls grasping, and the rotational movement of the fingers, which is translated into in-plane rotation of the end-effector. This combination of robust actuation and high-resolution sensing creates an intuitive and stable human-machine interface suitable for delicate manipulation tasks.

## 3.6 Cable Routing and Guidance System

### 3.6.1 Tip Design

In addition to motor performance, the accuracy and repeatability of the end-effector's motion are highly dependent on precise cable guidance and sustained tension control. To ensure consistent behavior during operation, each cable must be constrained to move freely yet securely within a meticulously designed path, minimizing unwanted friction and slack.

A specialized tip assembly has been developed to fulfill these requirements. As shown in Fig.3.3, the four cables—grouped into two pairs, each controlling one jaw of the plier—are routed through custom-machined channels. These channels ensure


**Figure 3.2:** Feetech 3009 for wrist inputs

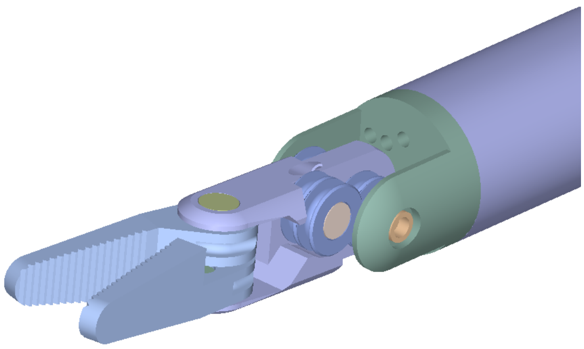

**Figure 3.3:** Tip Design for Cable Guidance

that the cables remain aligned throughout the entire range of motion, avoiding cross-interference and reducing wear.

The two jaw components (fabricated in light pink) are mounted onto a central middle base (green). To further enhance cable stability and minimize friction, donut-shaped guides machined via CNC (purple) are integrated into the middle base. These spheres act as low-friction directional pulleys, maintaining cable position with high precision and reducing bending stress, thereby extending cable service life.

The main base housing (pink) is designed with precision-aligned holes that direct the cables perpendicularly toward the motors. This alignment is critical for maximizing force transmission efficiency and eliminating off-axis loading, which can introduce positioning errors and reduce system responsiveness.

Through this integrated approach—combining dedicated guiding channels, spherical contact points, and perpendicular alignment—the system achieves high repeatability, minimal hysteresis, and smooth force feedback, making it suitable for delicate manipulation tasks requiring fine motor control.

### 3.6.2 Reel

A fundamental challenge in cable-driven mechanisms lies in maintaining adequate tension during both winding and unwinding phases. In a conventional single-motor-single-cable configuration, unidirectional actuation presents a significant limitation: while rotation in one direction (e.g., clockwise) effectively pulls and tensions the cable, reverse rotation (counterclockwise) does not positively push the cable but often results in slack accumulation along the path. This leads to unpredictable response, reduced positional accuracy, and potential entanglements.

To address this issue, a paired-cable design has been implemented. The two cables corresponding to a single degree of freedom are affixed to the same axial shaft driven by one motor. This configuration effectively reforms the two independent cables into a continuous loop—functioning conceptually as a flexible drive belt. As a result, when the motor rotates in one direction, one cable is pulled while the other is simultaneously recovered, enabling bidirectional force transmission without slack accumulation.

The key to ensuring effective operation of this belt-like linkage is sustained cable tension. To achieve this, the reel assembly is split into two distinct components, each dedicated to one cable. These two reel pieces are mounted on the same shaft and are designed to rotate in opposite directions relative to each other. This counter-rotation ensures that one cable is wound while the other is unwound during actuation, maintaining constant tension across the system.

To prevent slippage and undesired loosening of the cables during operation, one-way bearings are incorporated into the reel mechanism. As the name implies, these bearings permit free rotation in one direction while locking in the opposite direction. This feature is critical for preserving tension integrity, especially during transitions in

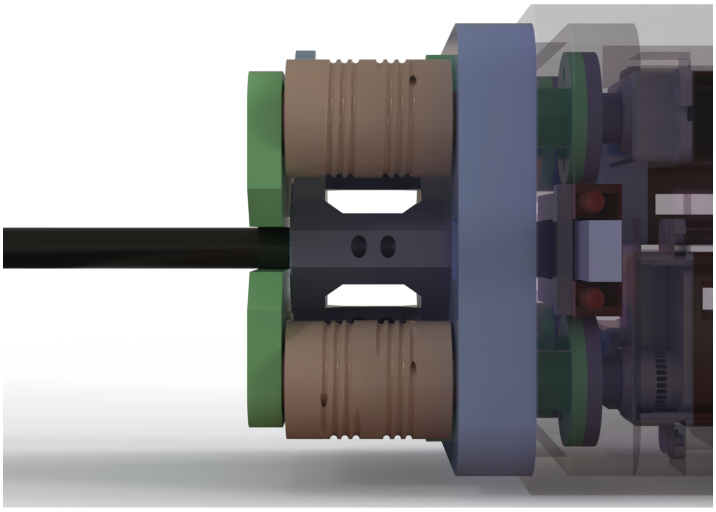

**Figure 3.4:** Reel Design and Winding Mechanism

motor direction or under varying load conditions. The use of one-way bearings enhances system reliability, improves motion accuracy, and minimizes the need for manual tension adjustments.

### 3.6.3 Motor Arrangement and Quick-Swap Mechanism

As previously discussed, the STS3032 servos are employed as the driving motors for the cable reels. In practical surgical scenarios, various end-effector tips are often required throughout a single procedure. To accommodate this need, a quick-swap system has been integrated into the design, allowing for rapid and secure tip replacement without compromising system integrity.

To achieve a compact form factor suitable for handheld operation, the four motors are arranged in a rotational symmetry pattern within the main housing. This layout optimizes spatial efficiency while maintaining sufficient clearance for motor power and communication cables, thereby preventing interference and ensuring reliable connectivity. The quick-swap mechanism incorporates a user-friendly yet secure attachment scheme. Press-activated hooks are mounted within the main frame, enabling tool-free engagement and release. To attach a tip module, the user simply aligns and applies gentle pressure until the hooks audibly lock into place. For disengagement, bilateral release buttons must be depressed simultaneously while withdrawing the module. This dual-action release strategy prevents accidental detachment during operation, enhancing safety, while minimizing the number of steps required for module exchange.

This design ensures a robust mechanical and electrical connection during use, maintaining precise cable alignment and tension integrity while facilitating efficient tool interchange in time-sensitive surgical environments.

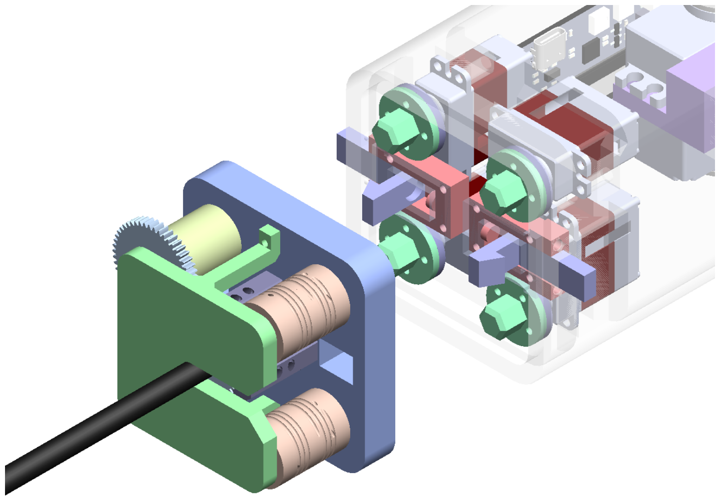

**Figure 3.5:** Motor Arrangement and Quick-Swap Mechanism

### 3.6.4 Input Design

The input module is designed to intuitively capture the user's hand and finger motions with high precision and minimal latency. By integrating two STS3009 servos and two AS5600 magnetic encoders, the system is capable of sensing four degrees of freedom (DoFs): two DoFs from the wrist movement—yaw and pitch—and two DoFs from the fingers, namely roll and open/close actuation.

The wrist motions are translated through a dual-axis gimbal-like mechanism, where the STS3009 servos provide both actuation force and preliminary positional feedback. Fine rotational movements in yaw and pitch are accurately detected and relayed to the control system, enabling responsive and natural manipulation of the end-effector. Finger actions are captured using the high-resolution AS5600 encoders, which detect both the rolling gesture and the degree of opening or closing of the fingers. Specifically, the open/close mechanism is implemented as an interchangeable module, allowing customization based on user preference or task requirements. The lever length and rebound force can be easily adjusted or swapped, facilitating ergonomic adaptability and reducing user fatigue during prolonged operations.

This modular and sensor-rich input design ensures that the system can accommodate a wide range of hand sizes and operating styles while maintaining accurate and reliable motion mapping between the user and the instrument.

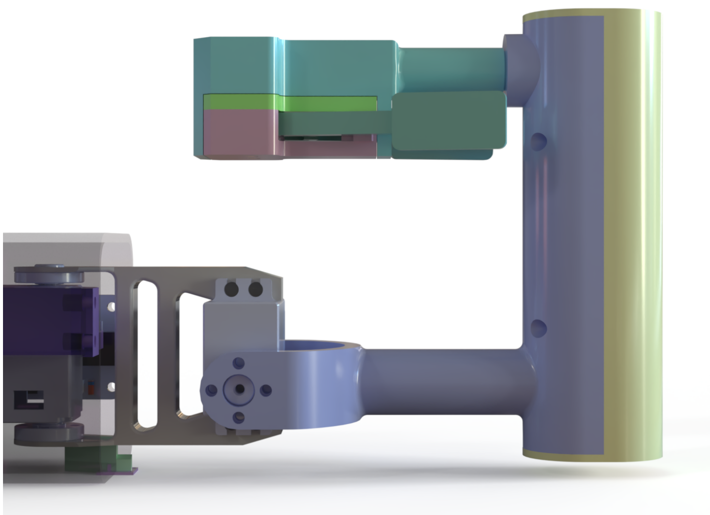

**Figure 3.6:** Input Module Design

## 3.7 Control Strategy

### 3.7.1 Digital Signal Logic

A critical aspect of ensuring precise control of the end-effector tip is the ability to accurately interpret user intent while rejecting extraneous inputs, such as physiological tremor or unintended vibrations. To achieve this, an advanced signal processing and control architecture has been implemented.

An Extended Kalman Filter (EKF) is employed to fuse and filter raw data acquired from multiple sources, including the AS5600 encoders and the feedback from the STS3009 servos. The EKF provides robust state estimation by dynamically weighting sensor inputs and predicting true user motion in the presence of noise and transient disturbances. This results in smooth, stable, and intentional control signals that faithfully represent operator commands while effectively suppressing high-frequency noise and involuntary motions.

At the core of the control system is an ESP32 microcontroller, which serves as the central computation unit. This chip is responsible for executing the EKF algorithm in real-time, processing sensory data, generating corresponding motor commands, and managing communication between system components. Its high processing power and support for floating-point operations make it well-suited for handling computationally intensive filtering and control tasks onboard.

After the user's input signals are filtered and interpreted, the refined commands are transmitted to the output-stage STS3032 motors that drive the cables. To further enhance motion accuracy and disturbance rejection, a PID control strategy is implemented based on real-time feedback from the STS3032 motors' built-in encoders. This closed-loop architecture enables precise regulation of cable tension and position, providing haptic transparency and ensuring safe, responsive operation during delicate surgical tasks. The combination of high-level EKF-based intention recognition and low-level PID motor control creates a robust hierarchical control system capable of compensating for both environmental uncertainties and human factors.

### 3.7.2 Human-Machine Interface

The control signals are transmitted from the rear input motors to the front output motors, ultimately driving the tip of the instrument. However, the system operates without physical switches or buttons to initiate or terminate control. To address this, a specially designed sequence of actions is implemented to ensure safe and intuitive operation.

Since the smart servos are equipped with absolute magnetic encoders, and their rotational range is limited to less than 360 degrees, the initial position of each motor can always be reliably identified. Upon powering on the device, the four front motors automatically execute a homing sequence. This process ensures that all cables are brought to a known tension state before any user input is accepted, thereby preventing uncontrolled movements and ensuring system readiness.

User control is enabled only after the homing process is complete. Slight pressure applied to the input lever serves as the trigger to activate the control linkage. This intentional design requires a deliberate action from the user, minimizing the risk of accidental engagement.

To terminate control, the user simply releases the lever. This action locks the rear input motors, effectively decoupling the user's hand from the instrument tip. This not only enhances safety by preventing unintended motions but also allows the entire instrument to be repositioned as a rigid unit during surgical procedures without affecting the end-effector's configuration. This approach provides a seamless and secure method for transitioning between active manipulation and passive positioning.

# Chapter 4

# Lasso Gripper: Design, Analysis, and Experiments

Building upon the research into the proposed medical instrument, the grasping capability of its end-effector relies fundamentally on the peak torque output of the motors and the maximum tension sustainable by the driving tendons. This conventional approach, however, encounters significant limitations when manipulating objects that exceed the maximum gripping distance or possess soft and fragile characteristics. In such scenarios, the demand for additional gripping force to achieve secure manipulation raises the critical risk of causing tissue damage or structural fracture in delicate biological organs.

To address this fundamental limitation in safe and adaptive grasping, this chapter introduces a novel gripper design. Lasso Gripper represents an integrated mechatronic system, strategically optimized for high-speed and high-load object manipulation while maintaining operational safety. The following sections will comprehensively elucidate its overall mechanical architecture, detail the specialized mechanisms for controlled string deployment and retraction, and present the associated control systems that enable its unique dynamic grasping functionality.

## 4.1 Design and Principle

Lasso Gripper comprises several crucial elements, including the robotic arm, controller, power supply, and communication interfaces. The robotic arm is engineered to provide the necessary degrees of freedom and precision required for accurate positioning and manipulation. The controller is responsible for managing the gripper's operations, processing sensor inputs, and executing the gripping actions. It employs advanced algorithms to ensure optimal performance. The power supply guarantees consistent electrical energy to all components, while communication interfaces enable seamless data exchange between the gripper and the central control system, ensuring synchronized operations.

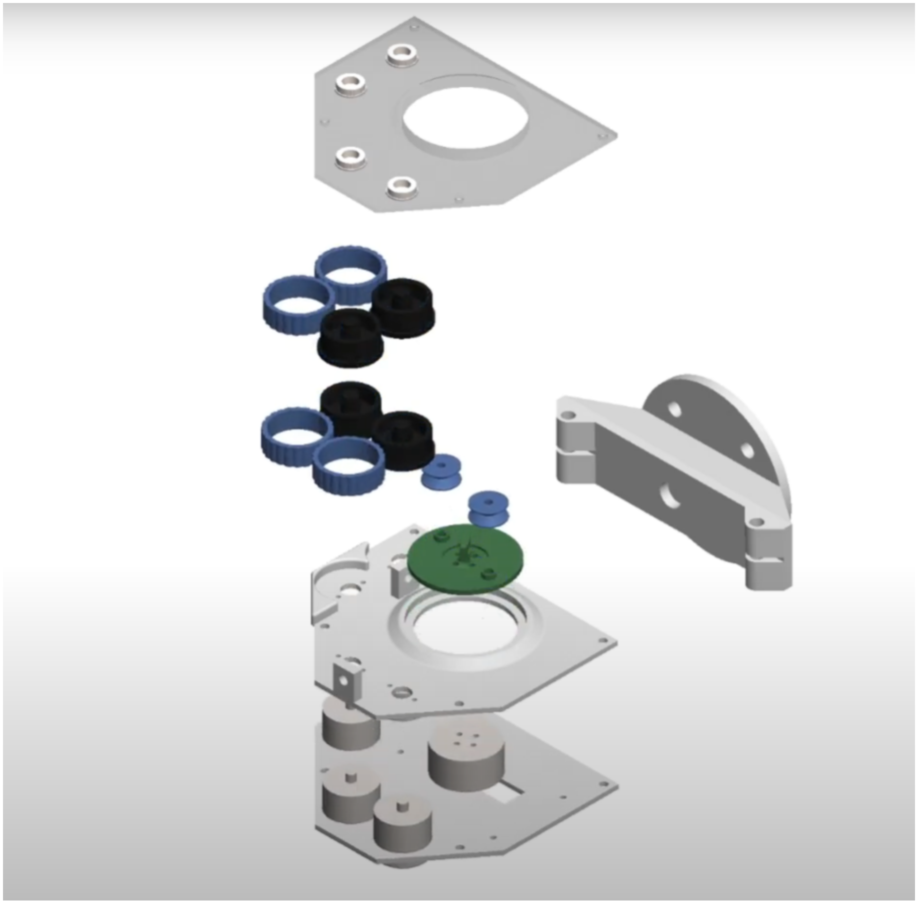

**Figure 4.1:** The SolidWorks designed hardware of the proposed gripper

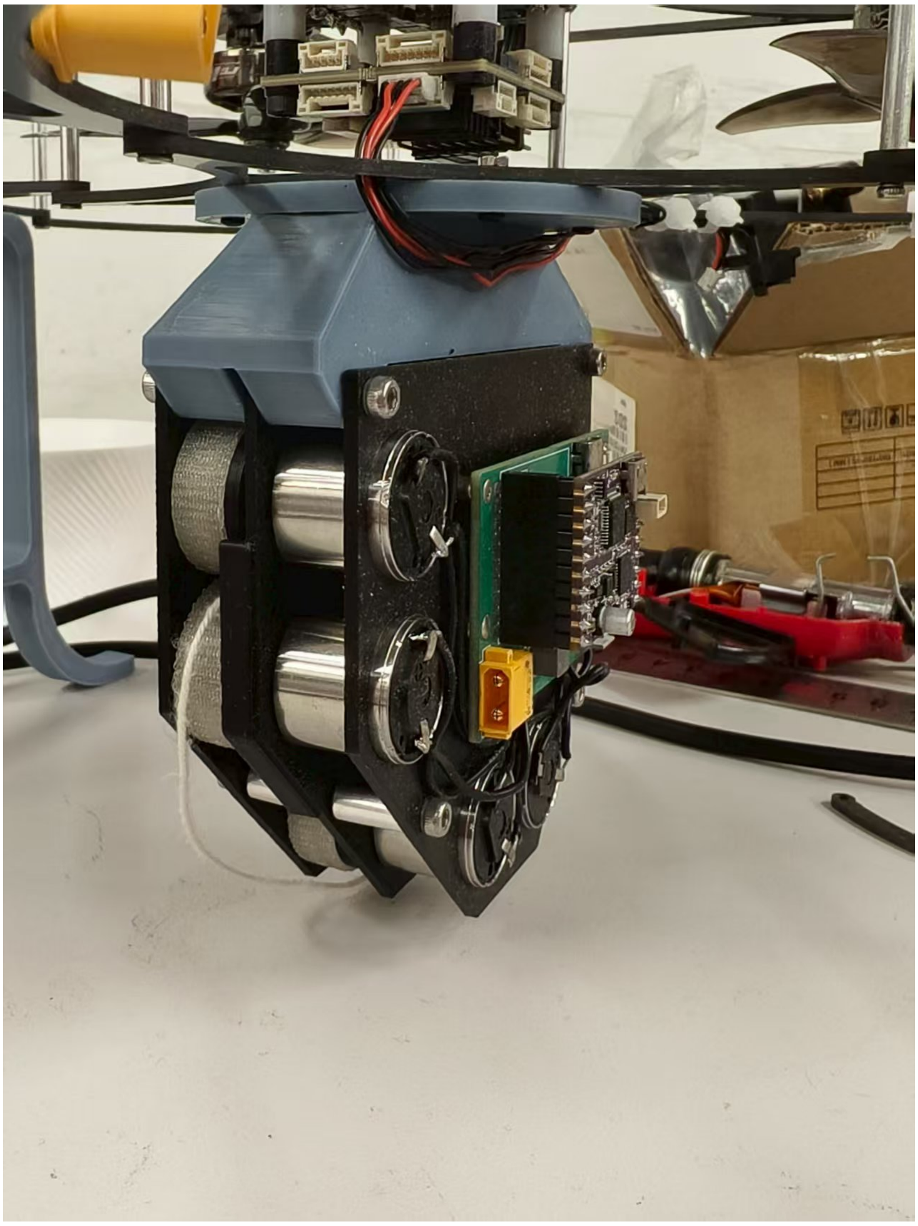

**Figure 4.2:** The proposed gripper connected with an ESP32 controller, motor electronic speed controller.

### 4.1.1 String Shooting and Retraction Mechanism

The string shooting and retraction mechanism is a defining feature of Lasso Gripper, enabling it to perform complex gripping tasks with high efficiency. The mechanism comprises multiple components working in unison to achieve precise control over the string's movement.

The propulsion of the string is achieved through two pairs of friction wheels. These wheels are designed with high-friction surfaces to grip the string effectively. The wheels are driven by coreless motors with high speed so that the required rotational speeds for rapid string extrusion can be achieved[23].

During the string shooting phase, both pairs of friction wheels rotate in the same direction, propelling the string forward through a guided track. This track is meticulously designed to ensure the string forms a loop within the desired dimensions and shape. Sensors along the track monitor the string's position and velocity, providing real-time feedback to the controller.

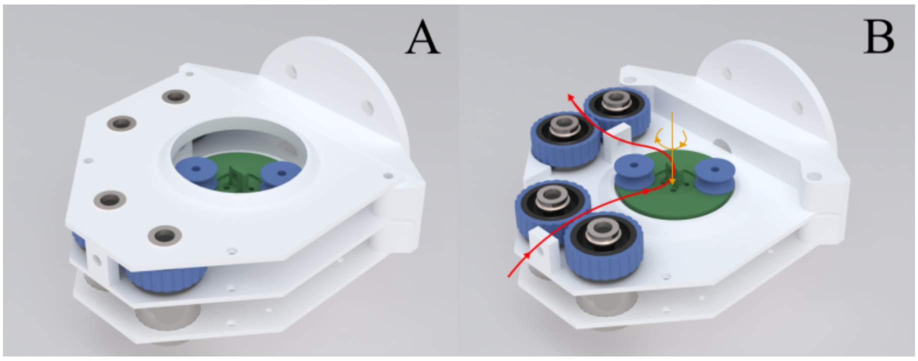


**Figure 4.3:** The friction wheels and the retraction mechanism driving motors are highlighted for illustration.

When retraction is required, the mechanism undergoes a coordinated sequence of actions. The rear pair of friction wheels maintains its rotational speed, continuing to pull the string inwards. Simultaneously, the front pair of wheels reverses direction, creating a counterforce that pulls the string loop into the space between the two pairs of wheels. This space is precisely designed to accommodate the retracted string without entanglement.

Within this confined space, a spool mechanism is designed. The spool is equipped with a rotational motor that winds the string onto itself, storing it neatly for future use. The tension exerted on the string during retraction is carefully controlled by a force feedback sensor located on the spool's axle. This sensor monitors the tension and adjusts the motor's torque to ensure that the object is gripped securely without applying excessive force that could cause damage.

### 4.1.2 The ESP32 Main Controller for Lasso Gripper

The ESP32 Main Controller serves as the central hub for managing the operations of Lasso Gripper. The ESP32 integrated Wi-Fi and Bluetooth capabilities, provides both the computational power and connectivity required for the gripper's complex functions. At the heart of the ESP32 Main Controller is its dual-core processor, which ef-

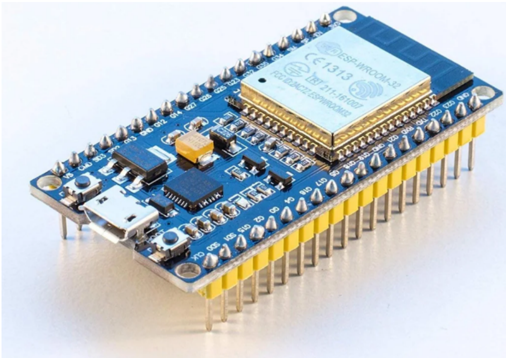

**Figure 4.4:** The ESP32 controller used in the proposed Lasso Gripper.

ficiently handles multiple tasks simultaneously. This includes real-time processing of sensor data, executing gripping algorithms, and maintaining communication with external devices. The controller interfaces with various sensors, such as force sensors and motion trackers, to gather critical data on gripping forces, positioning accuracy, and object integrity.

The ESP32's GPIO (General Purpose Input/Output) pins are utilized to control the retraction modules that adjust the grasping force of the lasso, ensuring precise manipulation of objects. Additionally, the controller's PWM (Pulse Width Modulation) capabilities allow for fine-tuned control of the friction wheels.

The integrated Wi-Fi and Bluetooth features of the ESP32 facilitate seamless communication with a upper computer for target recognition and grasping planning. This connectivity also allows for remote monitoring and control, data logging, and integration with broader automation systems.

## 4.2 Gripping Algorithm Design

The design of the gripping algorithm is critical to the functionality and efficiency of Lasso Gripper, ensuring precise and reliable manipulation of various objects. This section outlines the algorithm's design, which leverages advanced techniques for object recognition and manipulation, ensuring robust performance in diverse scenarios.

### 4.2.1

sectionObject Recognition and Point Cloud Processing The gripping algorithm begins with the recognition of the target object using an Intel REALSENSE depth camera D435i. The camera captures a 3D point cloud of the object, providing detailed spatial information about its shape and orientation. The point cloud data is processed to identify key features of the object, including its boundaries, surface characteristics, and potential gripping points.

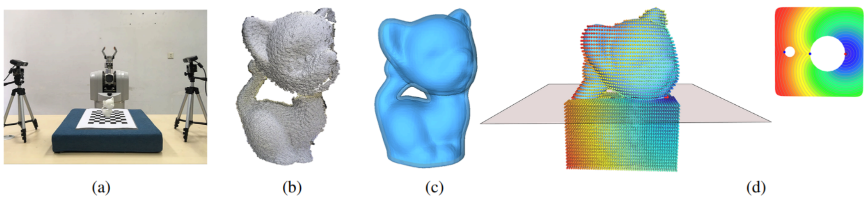


**Figure 4.5:** The caging loop calculating algorithm for Lasso Gripper [20]. (a) The setup of the depth reconstruction system (b)The Point cloud captured by depth camera directly. (c) A r-offset surface is built for defining the grasping space. (d) Two Morse saddle points (blue) are calculated to define the grasping loop.

The algorithm employs a method for identifying the "neck" of the object, which is a critical region for secure gripping. The method entails identifying caging loops, which are closed curves characterized within the embedding space of the target object's shape [20]. By decoupling the caging loops from the surface geometry, the algorithm can synthesize feasible caging grasps that are not constrained by the object's surface details. This enables the gripper to handle objects with complex geometries and topologies effectively.

### 4.2.2 Loop Positioning and Tightening

Once the neck region is identified, the algorithm determines the optimal position for the string loop around the neck. The positioning is crucial for ensuring that the loop can be tightened securely without slipping or causing damage to the object. The algorithm calculates the precise coordinates for the loop's placement, taking into account the object's shape and the desired gripping force.

The tightening process is executed in a controlled manner, with the string loop gradually constricting around the neck region. The wrist of the gripper approaches

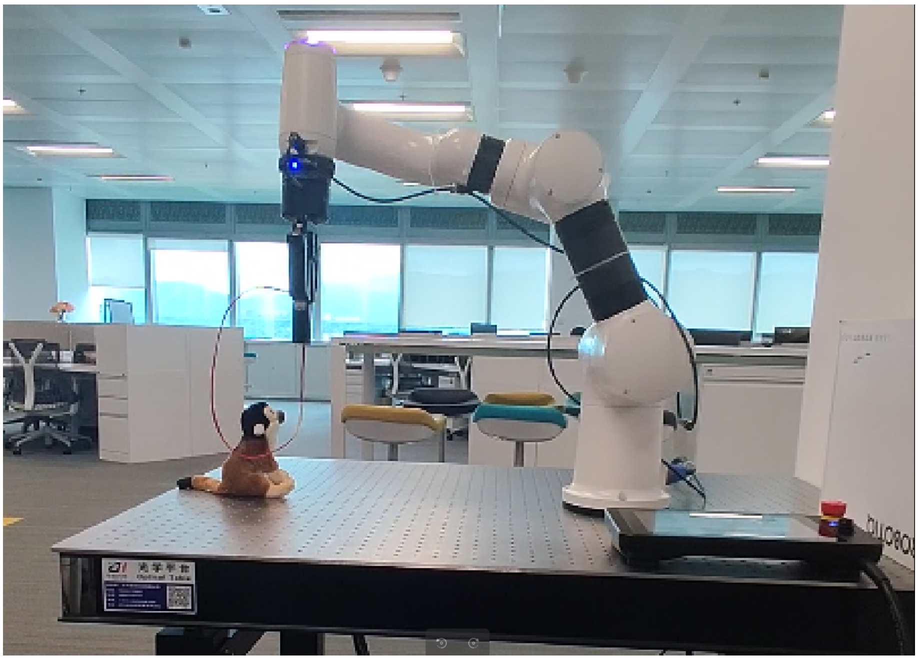

**Figure 4.6:** The illustration of the relative position of the string loop and the target object – an early test

the object in synchrony with the tightening action, ensuring that the loop maintains its position and grip. The following pseudo code illustrates the sequence of operations:

Input: Point Cloud P, String shooting velocity V

1. Scan P using depth camera
2. Process P to identify key features
3. Identify neck region N in P using the caging loop method
4. Determine optimal loop position L according to V
5. Align lower end of L with N
6. Execute controlled tightening action
7. Approach wrist towards object while maintaining loop position

### 4.2.3 Real-Time Feedback and Adjustment

The gripping algorithm incorporates real-time feedback mechanisms to adjust the gripping force and position dynamically. Sensors embedded in the gripper provide continuous data on the tension of the string, the position of the loop, and the force exerted on the object. This data is processed by the controller to make real-time adjustments, ensuring that the grip remains secure and stable.

The force feedback sensor, located on the spool's axle, plays a pivotal role in this process. It monitors the tension of the string during the tightening phase and adjusts the motor's torque accordingly. This prevents excessive force that could damage the object and ensures that the grip is firm enough to hold the object securely.

## 4.3 Dynamics of the String Loop

### 4.3.1 The Dynamics of String Analysis

The dynamics of the string loop in the lasso gripper are rooted in principles of fluid mechanics and physics [1][6]. This section explores the behavior of the string loop as it transitions from a low-velocity, gravity-dominated state to a high-velocity, self-supporting configuration.

At low velocities, the string loop is primarily influenced by gravity, causing it to hang downwards. This gravity-dominated regime is characterized by a noticeable sag in the string due to its light. As the velocity increases, the string undergoes a transition to a high-velocity regime where it forms a self-supporting loop. In this regime, the string is no longer suspended due to inertia but rather through the hydrodynamic drag. This drag force exerted by the surrounding air counteracts gravity, allowing the string to maintain a stable loop shape.

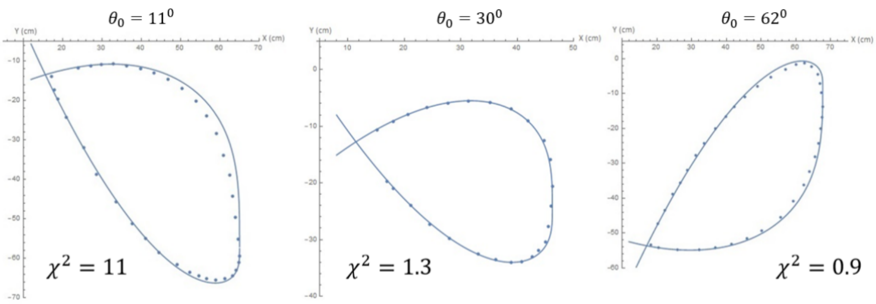


**Figure 4.7:** The string loop curve fitting of numerical theoretical solution with experimental profiles [1]. In this case, the linear mass $\mu = 0.283 \pm 0.005 g \cdot m^{-1}$, the length of the string loop $L = 160\,cm$, shooting velocity $v = 5.1\,m \cdot s^{-1}$ and sevel values of ejection pitch angle $\theta_0$. A satisfactory fitting result with $\chi^2 \leq 11$ is obtained.

Assuming the loop junction is at position $s = s_0$ when $t = 0$, the position of the junction along this curve is $s = v \cdot t + s_0 (\mathrm{mod}\, L)$. $T(s)$ is noted as the tension. Frenet-Serret frame $(\vec{\tau}, \vec{n})$ is used to describe the dynamics of an infinitesimal piece of string of length d$s$ subjected to its ight $\mu \mathrm{d}s\vec{g}$, its tension $\frac{\mathrm{d}(T\vec{\tau})}{\mathrm{d}s}\mathrm{d}s^3$ and the linear air drag $-f\vec{\tau}\mathrm{d}s$ where $f > 0$.

Newton's motion equation produces:

$$\frac{\mathrm{d}(\mu v \vec{\tau})}{\mathrm{d}t} = \mu \vec{g} + \frac{\mathrm{d}(T\vec{\tau})}{\mathrm{d}s} - f\vec{\tau} \tag{4.1}$$

Noting that $\mathrm{d}s = v \cdot \mathrm{d}t$, defining the effective tension $T^o = T - \mu v^2$, and recalling that $\vec{\tau} = \frac{\mathrm{d}\vec{\tau}}{\mathrm{d}s}$, the following can be obtained:

$$T^o\vec{\tau} - f\vec{\tau} + \mu\vec{g}s = \vec{0} \tag{4.2}$$

If the origin is wisely chosen for $\vec{r}$ as the point $o$ where the tangent to the loop is vertical, the integration constants will vanish. In the vertical local FrenetSerret frame, defining $tan\theta = \frac{\mathrm{d}z}{\mathrm{d}x}$, the following can obtained:

$$-f \cdot (x\tan\theta - z) + \mu g s = 0 \tag{4.3}$$

This integro-differential equation can be solved analytically as figure 4.1.

The formation of the self-supporting loop is governed by the hydrodynamic drag acting on the string. Daerr et al. (2019) investigated this drag both experimentally and theoretically, focusing on a smooth long cylinder moving along its axis [6]. The drag force is a result of the interaction beten the moving string and the air, creating a pressure differential that lifts and supports the string loop.

The equations that characterize the shape of the string loop re formulated under the assumption of an infinitely small string radius. These equations unveil a pivotal juncture, delineating two distinct regions: a supercritical region where the wave speed along the string is less than the string's own speed, and a subcritical region where the wave speed surpasses the string's speed. This critical juncture at the interface of these regions is vital to the string loop's dynamics. Solutions to the loop equations that remain regular at this critical juncture have been derived and subsequently compared to experimental data.

In practical terms, understanding the string loop dynamics has significant implications for the lasso gripper's design and operation. By leveraging the principles of hydrodynamic drag and the transition beten different velocity regimes, the lasso gripper can achieve precise control over the string loop's shape and stability. This enables the gripper to handle objects with varying sizes and shapes effectively, ensuring a secure and reliable grip.

## 4.4 The workspace analysis

Due to the self-supporting loop configuration of the Lasso Gripper in three-dimensional space, the operational workspace of the proposed manipulator significantly exceeds the physical reach of the robotic arm.

As delineated in Section 5.1, the three-dimensional pose kinematics of the lasso-type end-effector can be characterized by constrained dynamics under gravitational loading. For the local coordinate system constrained by gravity, the working plane

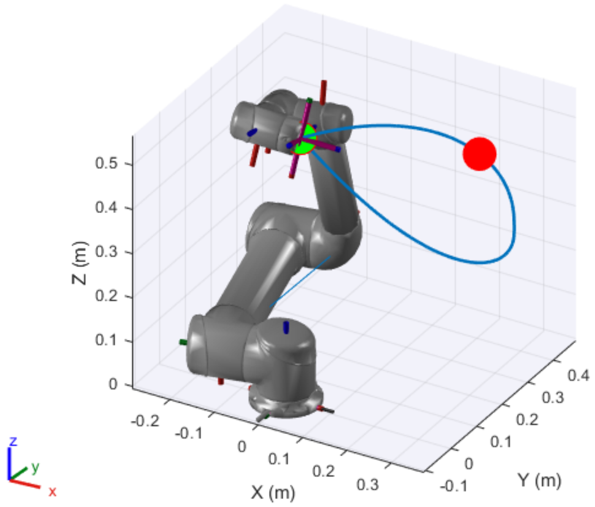


**Figure 4.8:** Lasso Gripper string loop is set at the end of UR5 robotic arm. The green point at the end of the robotic arm marks the position reachable by a rigid manipulator, while the red point indicates the furthest position on the string loop from the original point at the same robotic position.

is defined as the plane formed by the lasso's ejection direction and the gravity direction. The rotation with the normal vector of this working plane as the rotation axis is defined as pitch, while the rotation with the gravity vector as the rotation axis is defined as yaw. According to the right-hand coordinate system, the roll rotation axis is defined as the axis orthogonal to both the pitch and yaw rotation axes, lying within the working plane. Specifically, the gripper mechanism exhibits negligible roll angular displacement within the global coordinate frame due to gravitational stabilization. Its pitch orientation becomes parametrically dependent on the projectile's ejection velocity vector, while yaw angle variations are governed by the robotic manipulator's pose in Cartesian space.

A case study is conducted with specific parameters to investigate the working space of the Lasso Gripper on a 6-axis robotic arm using the Monte Carlo method. A UR5 robot without joint angle limitation is applied as an example in Fig. 4.8, and $[R, x_+, x_-]$ is set as $[0.7, 0.2m, 0.1m]$ in this case. The green point at the end of the robotic arm is the specific position that a rigid manipulator can reach, and the red point represents for the furthest point to the original point on the string loop at the same robotic position. The set of green points and red points in Fig. 4.9 is the workspace of the robotic arm and the extended lasso gripper, respectively. Generating 200,000 sampling points for robotic arm configuration, the volume of workspace convex hull is calculated. The volume of the original UR5 workspace convex hull is $3.4620m^3$ , and the volume of the extended Lasso Gripper workspace convex hull is $8.9002m^3$. The workspace extending ratio is up to 157.08%.

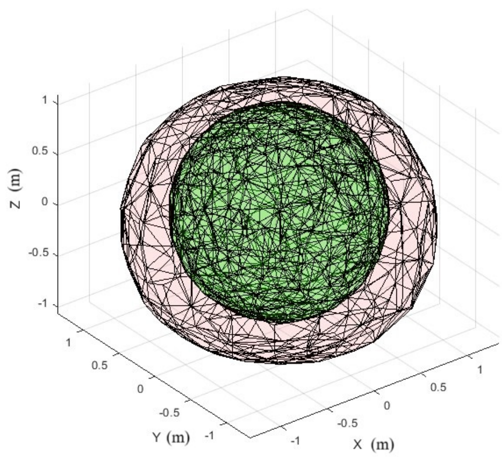


**Figure 4.9:** The initial robotic arm workspace is marked in green and extended Lasso Gripper workspace is marked in red.

## 4.5 Experimental Validation

### 4.5.1 Objectives

To validate the unique capabilities of Lasso Gripper and compare its performance with antipodal grippers, the following experimental procedure was designed. The setup employed a modular robotic arm (Inovo Robotic) with a maximum payload of 6 kg, mounted on a flat platform where the target objects were placed. The experimental objectives were divided into four categories:

1. Capturing animal figures by the neck or horns,
2. Capturing objects of diverse shapes,
3. Capturing oversized objects through shape adaptation, and
4. Capturing moving objects.

### 4.5.2 Experimental Setup

For Objectives 1 to 3, the procedure consisted of the following steps:

1. The robotic arm moved to a predefined position to initiate the launch phase;
2. Lasso Gripper was swept toward the target while maintaining its open-loop configuration;
3. Upon reaching the target, the string was retracted to secure the object;
4. The arm performed a vertical lifting motion while maintaining a stable grip.

For Objective 4, the robotic arm remained stationary, and Lasso Gripper independently executed the capture process aimed at intercepting a moving object in flight.

### 4.5.3 Experimental Procedure

The experimental procedure will be divided into several phases to thoroughly evaluate Lasso Gripper's performance:

1. **Static Object Gripping Tests**: In the second phase, static object gripping tests will be conducted to assess the gripper's ability to handle static objects. Test objects will be placed in fixed positions, and the gripping algorithm will be executed to secure and lift each object. Success rates, gripping forces, and any slippage or deformation of the objects will be meticulously recorded.
2. **Dynamic Object Gripping Tests**: The third phase will focus on dynamic object gripping tests to evaluate the gripper's performance with moving objects. Test objects with controlled motion, such as rolling or sliding, will be introduced, and the lasso will attempt to grip them. The effectiveness and speed of the gripping action will be measured to determine the gripper's capability in dynamic scenarios.

3. **Comparative Analysis**: Finally, a comparative analysis will be conducted to compare Lasso Gripper's performance with traditional gripping methods. Parallel tests using a standard parallel-jaw gripper will be carried out on the same set of test objects. Differences in gripping force, handling precision, success rates, and overall efficiency will be analyzed to highlight the advantages and potential areas for improvement of Lasso Gripper.

## 4.6 Data Collection and Analysis

Data collection will involve capturing both quantitative and qualitative metrics. Key metrics will include gripping force, measured using force sensors to determine the pressure exerted by the string loop on the objects, and positioning accuracy, assessed using motion tracking to evaluate the precision of loop placement around the object's neck. Success rates will be recorded as the percentage of successful grips out of total attempts. Additionally, object integrity will be inspected for any damage or deformation caused by the gripping process. Performance over time will be analyzed to identify any degradation in gripper functionality due to repeated use.

Statistical analysis will be performed on the collected data to identify trends, correlations, and areas for improvement. Comparative results will highlight the advantages and limitations of Lasso Gripper relative to traditional methods.

## 4.7 Results And Discussion

### 4.7.1 Various Objects Grasping Results

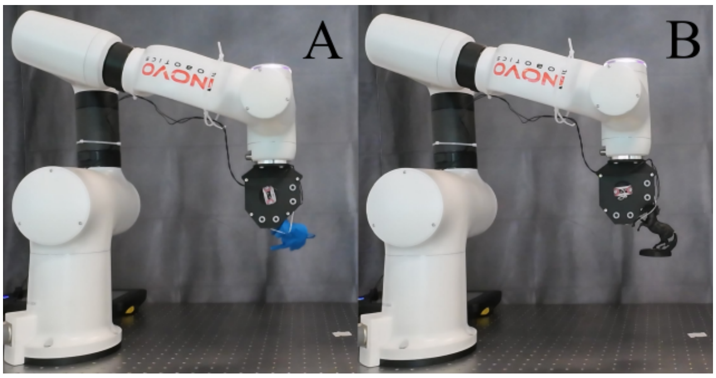


**Figure 4.10:** Validation of grasping ability via bull and horse figures.

Fig. 4.10 illustrates the experiments result for Aim 1. The lasso was successfully attached to the horn and the neck, respectively, and firmly grasped them towards the tip of the robotic arm, transporting the figures from the platform. For the second set of target objects, a drone, a cabbage, an empty coke can and a chili pepper were selected. In Fig. 4.11, Lasso Gripper successfully grasped the targets and carried it away from

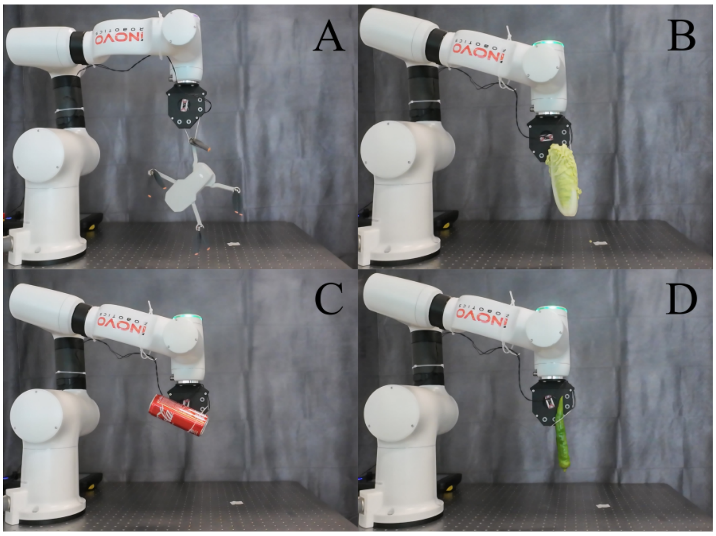


**Figure 4.11:** Validation of grasping ability over variable shapes: drone, cabbage, coke can and chili pepper.

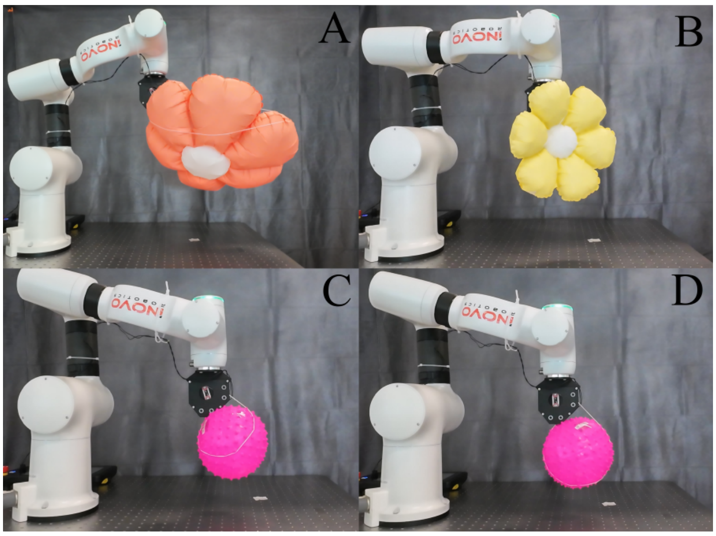


**Figure 4.12:** Validation of grasping ability with oversized objects: inflated balloons.

the base. The grasping strategy for these two experiments is similar. For both cases, the

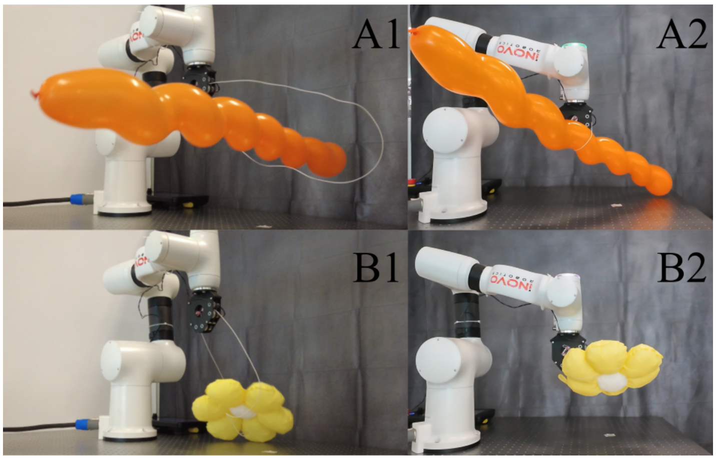


**Figure 4.13:** Flying object is captured by Lasso Gripper.

robotic swept right above the target while Lasso Gripper maintained the string in full range, ensuring that the target could be captured as the string retracted. This grasping strategy can also be extended to other expansive grasping targets, such as oversized balloons. Hence, the third set of objects are two flower-shaped balloons, with diameters of 45cm and 71cm, respectively, and a spike ball with a diameter of 18cm, as shown in Fig. 4.12. Though Lasso Gripper could not fully grasp the balloons due to the limitation in the string, it still firmly secured the balloons and lifted off the platform. This laid the foundation for the fourth case of the experiment, which was to capture flying objects. Fig. 4.13 presented the experiment by manually throwing one long balloon and the small flower balloon through the string made by Lasso Gripper.

Some interesting phenomena were observed during the demonstration, revealing more advantages of Lasso Gripper. Firstly, the string's sustained high speed makes it possible to grasp objects close to the table. In Fig. 4.11 (B) and Fig. 4.11 (D), where the cabbage and chili pepper are put very close to the table, Lasso Gripper can still maintain the loop shape and achieve the grasping task. The same goes for the demonstration near a wall. Secondly, the adaptive force closure of the string allows a certain degree of deviated grasping position to be tolerated. Compared with the grasping attitude in Fig. 4.12 (D), the string loop pose in Fig. 4.12 (C) occurs an unexpected deviation. Lasso Gripper realizes an adaptive force-closure grasping.

### 4.7.2 Comparisons with Antipodal Point Gripper

As the comparison group, the 2F-140 gripper from Robotiq was used as the classical antipodal gripper. In this scenario, the small flower balloon was used as the target due to its elasticity under external forces. As demonstrated in Fig. 4.14, the flower balloon was clearly under dramatic deformation as the antipodal gripper tried to pick it up from the platform. Though the balloon remained intact after the experiment instead of an explosion, the limitations of traditional grippers on handling delicate or plastic

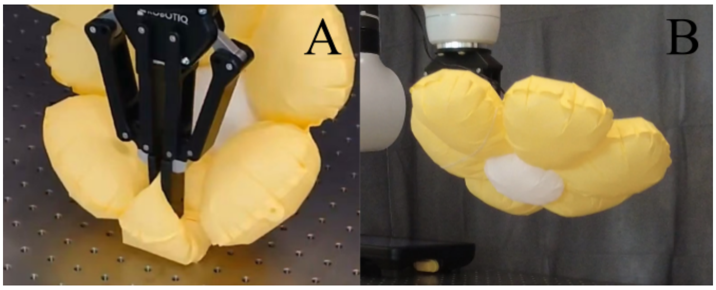


**Figure 4.14:** The inflated balloon is captured by the antipodal gripper. As a result of the concentrated stress applied, the balloon was severely deformed.

objects are straightforward. Instead of directly applying forces and pressure over the surface of the target, Lasso Gripper captured objects via rope tension, avoiding causing damage to the objects. Moreover, this unique feature enables Lasso Gripper to capture larger and potentially heavier objects than traditional grippers.

# Chapter 5

# Future Works

## 5.1 Surgical Instruments

While the current prototype of the cable-driven surgical instrument demonstrates robust functionality, there are significant opportunities for enhancement in precision, adaptability, and overall system intelligence. Future work will focus on the following two key areas:

### 5.1.1 Enhanced Precision through Adaptive Learning-based Control

A fundamental limitation of the current kinematic model is its assumption of fixed cable lengths. In practice, the stainless steel cables undergo microscopic plastic deformation and experience minute stretching after repeated loading cycles. Even a tiny elongation on the order of 0.1% can accumulate over the cable path, leading to a noticeable positional error at the end-effector—a phenomenon that compromises absolute accuracy and can manifest as a failure to reach a commanded pose.

To address this inherent model inaccuracy, a promising solution is to transition from a purely model-based control scheme to a data-driven, adaptive approach. We propose the integration of Reinforcement Learning (RL) algorithms. An RL agent can be trained, either in a simulated environment or through extensive real-world operation, to learn the subtle, non-linear relationship between motor commands and the resulting end-effector pose, effectively calibrating out the effects of cable stretch and hysteresis in real-time. By continuously correlating the commanded inputs with the actual achieved positions (potentially validated by external tracking systems during training), the model can self-correct and maintain high precision throughout the instrument's lifespan, moving beyond the limitations of a static kinematic model.

### 5.1.2 Expanded Modularity and Context-Aware Operation

The existing quick-swap mechanism provides a excellent foundation for hardware modularity. The next logical step is to exploit this capability by developing a diverse library of specialized end-effector modules (e.g., bipolar forceps, needle holders, scissors, and custom jaws for specific procedures). This would transform the instrument into a versatile multi-tool platform.

To fully realize the potential of this modularity, the system must possess the intelligence to automatically recognize an attached module and reconfigure its control logic accordingly. This can be achieved by embedding a Radio-Frequency Identification (RFID) tag within each module. Upon attachment, a reader in the main handle would instantly identify the module and load the corresponding kinematic profile and software limits (e.g., maximum safe grasping force for delicate tissues). For even greater flexibility and verification, a miniaturized computer vision module could be integrated near the distal end. This would allow the system not only to identify the tool but also to visually confirm its state (e.g., blade open/closed) and relative position to the surgical field, enabling advanced features like semi-autonomous tasks and enhanced safety monitoring.

Together, these advancements in intelligent control and context-aware modularity will pave the way for a new generation of surgical robots that are more precise, adaptable, and intuitive to use.

## 5.2 Further Applications of Lasso Gripper

Future investigations may involve utilizing multiple loops (two or three) to grasp the same object, potentially enhancing gripping stability and robustness. This multi-loop approach could provide additional security and adaptability in handling complex shapes and sizes, offering a promising avenue for further development.

### 5.2.1 Multi-Loop Configuration

Implementing a multi-loop configuration involves arranging two or more string loops to work in tandem. Each loop would be controlled independently but coordinated through a central control system to ensure synchronized movement and force application. This setup is particularly advantageous for gripping objects with irregular geometries or those requiring more secure handling.

### 5.2.2 Improved Gripping Stability

The use of multiple loops can significantly improve gripping stability. By distributing the gripping force across several contact points, the likelihood of object slippage or deformation is reduced. This is especially beneficial for delicate or fragile items that might be damaged by excessive pressure from a single loop.

### 5.2.3 Enhanced Adaptability

Multiple loops offer enhanced adaptability in handling a wide range of object sizes and shapes. Each loop can adjust its position and tension dynamically, conforming to the object's contours. This flexibility allows Lasso Gripper to manage complex tasks that single-loop systems might struggle with, such as picking up objects with protrusions or indentations.

### 5.2.4 Mobile Platform Grasping

As demonstrated in Chapter 6, Lasso Gripper successfully captured flying objects due to its rapid retraction capability. Essentially, this mechanism can be extended to applications involving object securing and transportation. With multiple loops, the gripper significantly increases the likelihood of catching and securing airborne objects.

### 5.2.5 Potential Challenges and Solutions

While the multi-loop approach offers clear advantages, it also introduces several challenges. Coordinating the movement and tension of multiple loops demands sophisticated control algorithms and real-time feedback systems. Additionally, the mechanical design must prevent interference among the loops during operation. Achieving the ultimate goal of capturing randomly flying objects further requires the grasping algorithm to be robust and adaptive to external obstacles such as walls or floors.

To address these challenges, future research should focus on developing advanced control algorithms that integrate sensor data for precise and dynamic adjustments. Moreover, the mechanical design could incorporate features to facilitate maintenance and quick reconfiguration, ensuring both reliability and operational efficiency.

# Chapter 6

# Conclusions

This study presents a systematic investigation into tendon-driven robotic systems, progressing from fundamental manipulation research to specialized surgical applications. The research demonstrates the versatility and adaptability of cable-driven mechanisms, establishing their significant value across both industrial and medical domains.

The work begins with the development of a novel handheld laparoscopic instrument that addresses critical limitations in minimally invasive surgery. This instrument incorporates a compact, modular architecture with a quick-swap end-effector system, achieving multiple degrees of freedom through precise control of four output cables. The integration of intelligent servos, high-resolution magnetic sensors, and advanced filtering algorithms ensures accurate translation of surgical intent while effectively mitigating physiological tremor, representing a significant advancement in surgical robotics.

Building upon the principles established in medical instrument design, the research further explores the fundamental aspects of tendon-driven manipulation through the development of the Lasso Gripper. This innovative grasping system employs a dynamic string loop mechanism that transitions from a gravity-dominated state to a self-supporting configuration, providing a flexible and robust solution for handling objects of various shapes and sizes. Experimental results demonstrate its superior performance over traditional antipodal grippers in manipulating complex geometries without causing damage, while maintaining capabilities for high-speed and high-load operations.

A comprehensive experimental framework validates both applications, confirming the effectiveness of the surgical instrument in enhancing surgical dexterity and the Lasso Gripper's proficiency in grasping both static and dynamic objects. These parallel developments highlight the cross-domain potential of tendon-driven systems.

Looking forward, two key research directions emerge from this work. First, the integration of data-driven approaches such as Reinforcement Learning can address inherent system nonlinearities, including cable stretch and hysteresis, to achieve higher precision control. Second, the development of intelligent, auto-identifying modules through technologies like RFID will enable greater versatility and context-aware operation, particularly beneficial for surgical applications.

In conclusion, this research validates the transformative potential of tendon-driven mechanisms through both medical and industrial applications. By bridging fundamental robotic principles with practical clinical and manipulation needs, this work establishes a foundation for future innovations in intelligent, adaptive robotic systems that span healthcare and industrial domains.

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